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Bayesian Tools of the Trade for Developmental Psychologists: A Quick-Start Guide Using JASP

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ABSTRACT

Bayesian methods offer many advantages when applied to psychological research, yet they may seem esoteric to researchers who are accustomed to traditional methods. This paper aims to lower the barrier of entry for developmental psychologists who are interested in using Bayesian methods. We provide worked examples of how to analyze common study designs, demonstrating how to use Bayesian versions of traditional statistical routines (t-tests, binomial tests, correlation, linear regression, and ANOVA). By adopting this approach, we focus on providing researchers with Bayesian analyses that will be both immediately applicable and somewhat familiar. In the process, we introduce fundamental Bayesian concepts and fundamental challenges that come with using Bayesian methods.

Introduction

Bayesian methods have been growing in popularity over the past decade, in large part because they are remarkably convenient. For developmentalists, Bayesian statistics can offer particularly attractive benefits. First, researchers can quantify evidence in favor of the null hypothesis (Wagenmakers et al., 2018). This feature is valuable for all scientists; for developmental researchers, it enables them to, for example, find evidence that an effect does not differ between age groups. Second, researchers can start collecting data without a clear sense of their ultimate sample size. Recruiting children as participants is challenging, so it can be quite helpful for researchers to be able to adjust their sample size on the fly. They can look at results while still collecting data and stop once they have sufficient evidence or keep recruiting participants until they can be confident in their conclusions. All of these actions would be more difficult using the traditional frequentist Null Hypothesis Significance Testing (NHST) approach,[1] but are automatic under a Bayesian approach (Rouder, 2014). Bayesian methods are also less conceptually challenging than they might seem. Indeed, the results are actually easier to interpret than those of an NHST analysis. One primary output of a Bayesian analysis is a Bayes factor. This directly quantifies how much more likely the data are given one hypothesis than given another. In contrast, NHST analyses produce p-values, which quantify how likely we are to obtain data as extreme or

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[1]For a thorough comparison of Bayesian and frequentist statistics, we refer readers to (Dienes, 2011)

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more extreme than the data we have, assuming that our null hypothesis is correct. The Bayes factor is the information we want in an easily digestible format.

If Bayesian methods are so intuitive, why weren't Bayesian methods standard for the past 50 years? The answer is that the computations involved in Bayesian methods can be rather intricate. Even now, beyond intentionally simple examples, Bayesian methods necessitate the use of specialized software (e.g., JAGS; Plummer, 2003), which employ sampling algorithms such as Markov chain Monte Carlo (MCMC; Van Ravenzwaaij et al., 2018). Today, many developmental psychologists prefer statistical routines like t-tests simply because they are familiar. Researchers who are already using Bayesian methods often adopt more free-form approaches, creating an impression that one must be proficient with customized models to implement Bayesian methods. Yet there are Bayesian routines comparable to the traditional frequentist null hypothesis routines that many developmental psychologists rely on. With the introduction of the GUI-based JASP software (JASP Team, 2023), which implements Bayesian analysis and sampling algorithms in an SPSS-style framework, the barrier to entry for using those Bayesian routines has been lowered considerably. In this article, our aim is to lower it still further by providing a practical tutorial of how to apply such Bayesian methods to analyzing common study designs in developmental psychology. Our hope is that when developmental scientists get comfortable using Bayesian methods to do familiar things like t-tests and ANOVAs, they will begin to explore how Bayesian methods can be used to do many other things which are not possible in a traditional frequentist framework.

This article should be thought of as a quick start guide rather than a comprehensive account of Bayesian methods. As with any methodology, there are important debates about how best to conduct Bayesian analyses. We encourage enthusiastic readers to further explore the nuances of Bayesian analysis, and we provide citations throughout to facilitate such exploration.

We begin with a worked example of a Bayesian t-test. We recommend that all readers read this first case study before moving on to other analyses because we use this first case study to introduce the fundamental principles of Bayesian analysis using a concrete example. We then expand on these principles in additional worked examples, demonstrating a Bayesian approach to analyzing a two-alternative, forced-choice experiment, a correlational study, and a basic factorial design. These subsequent case studies are intended for *à la carte* consumption and thus may contain repetitive elements if the paper is read from start to finish. Finally, we highlight general considerations for using Bayesian analyses.

Case study 1: t-test and introduction to Bayes

For our first worked example, we use data from an experiment conducted by Thomas et al. (2022). Specifically, we use data from Thomas et al.'s Experiment 2A. In this experiment, researchers were investigating how toddlers make inferences about the relative strength of social relationships. The researchers suspected that toddlers might interpret acts that demonstrate a willingness to exchange saliva (such as in sharing food) as indicating particularly "thick" social relationships.

To examine this possibility, Thomas et al. used a preferential looking time paradigm, presenting participants ($n = 27$; 16.5–18.5 months old) with a video of a puppet flanked by two actors. First, participants were familiarized with the actors' willingness to interact with the puppet in a food-sharing or non-food-sharing way. One actor shared an orange slice with the puppet, while the other actor passed a ball back and forth with the puppet. In the test phase, the puppet expressed distress, and the researchers recorded how long participants looked at each actor.

The researchers interpreted a tendency to look longer at a particular actor upon seeing the puppet exhibiting distress as indicating that the participant expected that actor to respond to the puppet's distress, which would in turn indicate that participants had inferred a thick social relationship between the puppet and that actor. They hypothesized that, when the puppet exhibited distress, participants would look longer at the actor who had shared food with the puppet.

The dependent variable for this experiment was calculated as

$$\frac{\text{time looking at food actor}}{\text{time looking at food actor} + \text{time looking at ball actor}} \tag{1}$$

A value of 0.5 indicates that participants looked at each actor equally. A value of 1 indicates that participants only ever looked at the food actor, and a value of 0 indicates that they only ever looked at the ball actor. The null hypothesis was that participants look equally at each actor on average; the alternative hypothesis was that participants look longer at one actor or the other. More formally, we can say the null hypothesis is that the population mean of the dependent variable equals 0.5, indicating no tendency to look at either actor more than the other. In mathematical terms, we can write this as $H_0 : \mu = 0.5$. The alternative hypothesis is that the population mean equals something other than 0.5, indicating a tendency to spend more time looking at one of the actors. We can write this hypothesis as $H_1 : \mu \neq 0.5$.

Analysis approach

We have now characterized the hypothesis of Thomas et al. in terms of a statement about the population mean's value. We want to measure looking times in our sample and then use our data to infer whether the population mean of the dependent variable is 0.5, as specified by the null hypothesis. We can make that inference using a one-sample t-test. But before focusing on our specific example, we must first consider how to approach Bayesian t-tests in general.

In the Bayesian approach to conducting a t-test, we focus on making inferences about the population effect size.[2] By focusing on standardized effect sizes, we can have a statistical routine that can readily be applied to any relevant context, regardless of the dependent variable's scale or units. To do so, we reframe our hypothesis test so that the null hypothesis corresponds to saying that the population effect size is zero, because such an effect size indicates that participants are looking at both actors equally (i.e., participants do not expect either actor to respond more than the other to the puppet's distress). If the population effect size is positive, then there is a tendency to look longer at the actor who shared food with the puppet. In the context of the present experiment, we would interpret such an effect size as

[2]The sample effect size is defined as $d = (\bar{y} - \mu_0)/s$, where $\bar{y}$ is the sample mean, μ_0 is the test value, and s is the sample standard deviation. The population effect size is $\delta = (\mu - \mu_0)/\sigma$, where μ is the population mean and σ is the population standard deviation. In the present case, our μ_0 test value is 0.5.

Box 1: Bayes' Rule.

Bayes' rule can be written as

$$p(\theta|\mathrm{y}) = \frac{p(\mathrm{y}|\theta)p(\theta)}{p(\mathrm{y})} \tag{2}$$

where y represents our data and θ represents a parameter that we wish to estimate. As researchers, we may consider certain values of the parameter of interest to be theoretically meaningful. For example, if our parameter of interest is effect size, our null hypothesis proposes that the parameter of interest is 0.

The $p(\theta|y)$ notation in Bayes" rule denotes the probability of θ given y. Since θ is our parameter of interest and y is our data, Bayes' rule is showing us how to calculate the probability of a parameter of interest taking any possible value given our data. Since it constitutes our belief *after* having seen our data, $p(\theta|y)$ is commonly referred to as our posterior probability. The posterior probability distribution represents our relative beliefs for all possible values of our parameter of interest. This distribution tells us the relative plausibility of each parameter value within its range of possible values. To calculate the posterior probability, Bayes' rule tells us that we need to account for three terms. The first is $p(\theta)$, our belief about the value of the parameter of interest (in this case, the effect size) before we consider the current data. This term is commonly referred to as the prior distribution, as it represents our beliefs *before* we had the current data. Hence, inference accomplished with Bayes' rule is often referred to as Bayesian updating; we use our data to update what we initially believed.

The second is $p(y|\theta)$, or the probability of collecting such data given the parameter of interest (i.e., assuming that the parameter takes a specific value). Intuitively, this term considers the predictions made by each hypothesis: if we knew the true value of our parameter, we could calculate the probability of collecting the data that we have. For example, if we assume the data are normally distributed, then each value of the mean parameter would imply a different normal distribution for the data. Data points toward the middle (i.e., near the mean) of each distribution would be considered more likely to be observed, given that proposed mean. In this article, we explain how to craft such hypotheses in the context of each worked example.

The remaining term, $p(y)$, is the marginal probability of the data. This term is essentially the average of all $p(y|\theta)$ weighted by the prior distribution $p(\theta)$. This component is important to a deeper understanding of Bayesian model comparison and is centrally involved in the underlying computations that produce those results. We encourage interested readers to refer to Rouder and Morey (2019) for an overview. Yet as we will see, it is not a crucial component for interpreting or reporting results under the basic approach we outline in this article.

Once we have formalized our hypothesis and specified a prior distribution, JASP can compute the posterior distribution, $p(\theta|y)$. It does so using various computational algorithms that are beyond the purview of this paper, but ultimately, each one is a clever way to take integrals of potentially complex mathematical functions.

indicating that participants had inferred a relatively "thick" social relationship between the puppet and the food actor.

In a Bayesian context, we use probability to quantify the uncertainty we have about the state of the world. In this example, we do not know the true value of the population effect size (i.e., we don't know whether toddlers as a population would look longer at the food actor or not), but with Bayesian methods, we can quantify what we do know about it – what we *believe*. Our initial beliefs carry more weight in a Bayesian framework than one might assume based on how the word "belief" is ordinarily used. In a Bayesian context, our beliefs are not arbitrary. Instead, they may be characterized as a theoretical position based on whatever evidence we have so far, be it past experiments or general life experience. Our beliefs are thus inherently tentative but represent our best currently-available knowledge. Bayesian methods then allow us to take into account the data that we have collected in the present experiment and determine how our best available knowledge should change. In other words, we can use these methods to quantify how an experiment has altered our knowledge. This philosophical angle can strike some researchers as odd, so we return to it when we discuss the importance of our prior beliefs in the next section ("Specifying Priors").

Since the population effect size could be any real number, we quantify our beliefs using probability distributions. For examples of probability distributions, see Figure 4, which shows two probability distributions, both over possible effect sizes (our parameter of interest in a Bayesian t-test). The height of the distribution at any given point tells us the relative plausibility (i.e., how strongly we should believe) of each possible value of the population effect size.

We update our beliefs about the population effect size using Bayes' Rule, which gives us a principled method for deciding what to believe about a parameter of interest after taking into account our data. JASP takes care of the computations, but below we provide a broad explanation of the inner workings of Bayes' Rule in an inset. For those who are interested in greater detail, there are numerous tutorials discussing the nuances of Bayes' rule (e.g., Etz & Vandekerckhove, 2018; Lindley, 1993).

There are three main takeaways from Box 1. The first is that the output of Bayes' Rule is a probability distribution representing what we believe about a parameter of interest, given the data we have collected. With our parameter of interest for a t-test being the effect size, this posterior probability is precisely what we want to know: What should we believe about our effect size, given these new data?

The second takeaway is that, for Bayes' Rule to work, we need to specify our hypotheses in a form that would allow us to calculate the probability of our data, assuming we know the true value of a parameter of interest. In our example, we have already done this by specifying our hypotheses in terms of standardized effect sizes and in the assumptions we make by choosing a t-test.

The third takeaway is that, beyond specifying our hypotheses and providing data, all JASP requires from us is a probability distribution that represents our beliefs prior to an experiment. We devote the next section to the idea of prior probability, as it is a common sticking point for people who are new to Bayesian methods.

Specifying priors

The basic mechanism of Bayesian inference is that we use our data to update our prior beliefs, producing our posterior beliefs. Our prior beliefs serve as our starting point, so as the first step to any Bayesian analysis, we need to explicitly state what our prior beliefs are. In many cases, we want to make as few *a priori* commitments as possible. As researchers, we have an understandable desire to remain neutral regarding our hypotheses; we want the data to speak for themselves. Prior distributions that incorporate only minimal background knowledge are often called "uninformed" priors, and because they make minimal theoretical commitments, they can be used in a variety of circumstances. Uninformed priors thus make for excellent defaults, and the software we use in this tutorial (JASP; JASP Team, 2023) does indeed default to such priors.

Yet even with uninformed priors, we are still responsible for making sure the priors are sensible in the context where we are using them. Making minimal theoretical commitments is itself a theoretical commitment. It is thus important to understand exactly what we are saying about our beliefs when we choose to use uninformed priors.

It can be surprisingly difficult to construct sensible uninformed priors (in the case of t-tests, see Rouder et al., 2009). To illustrate the challenge, consider how we might try to specify our prior beliefs regarding a t-test. Our first instinct might be to describe our expectations about the group means and standard deviations. Yet such an approach would require us to use our specific experimental context to inform our priors, since we would need to situate our expectations on the scale of our dependent variable. For example, how long a baby looks at a stimulus in milliseconds is on a completely different scale than how many stickers a preschooler is willing to share with a peer. To avoid making such experiment-specific judgments, we can instead think about our priors in terms of what standardized effect sizes we might expect to see. This is why we focus on the standardized effect size as our parameter of interest.

Having shifted our focus to effect sizes before running our experiment, we still face the challenge of how to avoid making commitments about what we expect. We might want to say that we think all outcomes are equally likely, which we could do by specifying our prior as a uniform distribution over all possible values of the effect size. This approach might at first seem intuitively appealing because we avoid favoring any particular value over any other value. Yet rather than being noncommittal, specifying such a prior would actually mean committing to ridiculous theoretical positions. For example, we would be saying that we think the effect size is just as likely to be between 0 and 10 as it is to be between 100 and 110. To put that in perspective, psychologists typically treat effect sizes of 0.2, 0.5, and 0.8 as small, medium, and large effects, respectively (Cohen, 1992). If we tried to eliminate absurdly high values by making the range finite, say by specifying our priors as a uniform distribution between −4 and 4, the beliefs expressed by a uniform prior would still not make sense in most contexts. First, we would be saying that we think an effect size of 4 is just as likely as a more moderate effect size of, say, 0.5. Second, we would be saying that we are absolutely certain that the effect size could not be greater than 4 or less than −4. Such a statement does not reflect our beliefs in most cases, as we know that such extreme effect sizes are possible, just unlikely.

Attempting to make absolutely no commitment is a fool's errand. Instead, we are looking for priors that make *minimal* commitments while avoiding nonsensical theoretical positions. It is important to be aware of the commitments we do end up making in adopting an uninformed prior so that we make adjustments when those commitments are inappropriate for a given situation.

A popular option for t-tests, and the prior that JASP defaults to, says that we think smaller effect sizes are generally more likely than larger effect sizes. Specifically, we represent our prior beliefs with a Cauchy distribution with a location of 0 and a scale of 0.707 (dashed line in Figure 4). This prior has the benefit of matching with our intuitions about typical effect sizes while also being statistically convenient in a variety of ways that are beyond the scope of the present tutorial.[3] Suffice it to say that we use this prior as a default uninformed prior for t-tests because it aligns with our typical expectations, avoids incorporating any information specific to a given experiment, andallows our data to be the driving force behind our conclusions.

While uninformed priors are a decent default, they are not always the best choice. We often conduct experiments because we at least have some vague suspicions about how the results might turn out. If we fail to incorporate such suspicions into our prior probability, then we misrepresent how our beliefs have changed as a consequence of our data. JASP allows us to make relatively minor adjustments to our priors by retaining the overall shape of the Cauchy distribution and altering the parameter values. The location parameter specifies the center of the distribution, while the shape parameter specifies its spread. We can change the location of the Cauchy if we think that positive or negative effects are more likely. The default location of zero means that we are agnostic about the direction of the effect. We can also increase or decrease the scale if we think extreme effects are respectively more or less likely. Greater customization is also possible, and there are tutorials for how to incorporate specific knowledge about how participants are likely to behave (Christensen et al., 2010) as well as vaguer knowledge about the likely size of effects within a domain of study (Gronau et al., 2019).

Researchers who are new to the Bayesian approach to statistical analysis are sometimes uneasy about how much importance is placed on what we believe about an effect before we see our data. For one thing, this prior probability is inherently subjective. We have to choose our priors, and different people may make different choices based on their own experiences. Our prior beliefs might be informed by past experiments, but we would still be making a judgment about the applicability of past experiments to the present experiment. In thinking about this concern, it may be helpful to approach our choice of priors as deciding what assumptions we are willing to make. Even ostensibly objective analyses under the classical approach rely on us deciding what assumptions we are willing to make. For example, simply choosing to use a t-test involves assuming that the data are normally distributed and that error is independent for each data point. The Bayesian approach involves making assumptions about what we think the results of an experiment could reasonably look like. JASP also offers a simple option for examining how different priors would impact our results.

Inferential outputs

Once we have specified our prior distribution, JASP can use our data to update it, producing our posterior distribution. The posterior represents what we now believe about the parameter of interest, after doing our experiment and considering our data. However, since the posterior distribution is a probability distribution, it does not provide a readily communicable summary in itself. Instead, it provides the relative plausibility of parameter values. We need statistics that highlight our key conclusions.

In thinking about what statistics would be helpful, we are interested in two considerations. The first is completing the hypothesis test: do our beliefs about the parameter of interest align more with the null or alternative hypothesis? In other words, is there evidence

[3]Further inspetion reveals that this prior has the desirable trait of having minimal impact on the ultimate shape of the posterior distribution—less than that of a single data point.

for or against an effect? The second is estimating the value of the parameter of interest itself: if there is an effect, what is its direction and magnitude?

To report the results of a hypothesis test, the most commonly used statistic is the Bayes factor. A Bayes factor is a comparative measure of the evidence provided by our data. Specifically, the Bayes factor tells us how the balance of evidence between two competing hypotheses shifts after we use our data to update our beliefs. It quantifies this balance of evidence in terms of a ratio of two probabilities: the probability of our data given the null hypothesis and the probability of our data given the alternative hypothesis. Formally, a Bayes factor in favor of the alternative hypothesis is written as $BF_{10} = P\ (\text{data}|H_1)/P\ (\text{data}|H_0)$.

Since BF_{10} is saying the data are that many times as likely to occur under the alternative hypothesis as under the null, $BF_{10} = 1$ indicates that our data are equally probable under either hypothesis. $BF_{10} > 1$ indicates that our data are more likely to occur if the alternative hypothesis is correct, and $BF_{10} < 1$ indicates that our data are less likely to occur if the alternative hypothesis is correct.

Note that since a Bayes factor is a ratio, we could also specify a Bayes factor in favor of the null hypothesis: $BF_{01} = P\ (\text{data}|H_0)/P\ (\text{data}|H_1)$. Since BF_{01} is just the inverse of BF_{10} (and vice versa), we can readily convert one output to the other. It is typically easier for readers to interpret numbers greater than 1 rather than decimals or fractions, so we recommend selecting BF_{01} or BF_{10} based on which will provide values greater than 1, although this is not strictly necessary. For example, if $BF_{01} = 0.091$, we could say that our data are 0.091 times as likely to occur under the null hypothesis than under the alternative hypothesis, or we could say that our data are 11 times as likely to occur under the alternative hypothesis than under the null hypothesis. We suspect that the latter is much easier for most people to interpret.

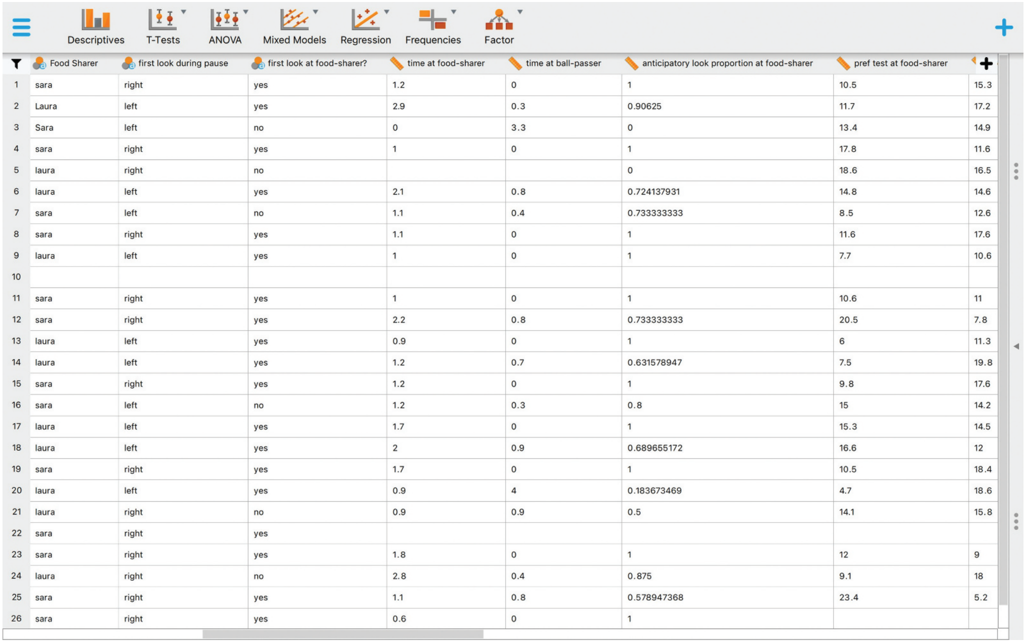
To estimate the value of the parameter of interest, the most commonly used summary of the posterior distribution is the central 95% credible interval. We can be 95% certain that the parameter of interest has a value that lies within the range of its 95% credible interval. In the case of a t-test, credible intervals can help us determine the likely direction and magnitude of a difference. We could also theoretically provide a point estimate of the parameter, such as the mean, median, or mode of the posterior distribution. Yet the 95% credible interval is particularly important because it takes into account our uncertainty about the value of the parameter, so it tells us a range of values that we should consider plausible.

To summarize, we will be using Bayes factors as our primary outputs for hypothesis tests and credible intervals as our primary outputs for parameter estimation throughout this tutorial. For this first case study, Bayesian t-tests focus on whether data provide more support for the null hypothesis (that there is no effect) or the alternative hypothesis (that there is an effect). We use Bayes factors to quantify how much more probable our data are under one of these hypotheses. We use credible intervals to quantify the likely magnitude and direction of any effect.

Worked example

Initial steps

Having discussed the Bayesian approach to t-tests in general terms, we will now walk through a concrete example of a Bayesian one-sample t-test using the data from Thomas



	Food Sharer	first look during pause	first look at food-sharer?	time at food-sharer	time at ball-passer	anticipatory look proportion at food-sharer	pref test at food-sharer	
1	sara	right	yes	1.2	0	1	10.5	15.3
2	Laura	left	yes	2.9	0.3	0.90625	11.7	17.2
3	Sara	left	no	0	3.3	0	13.4	14.9
4	sara	right	yes	1	0	1	17.8	11.6
5	laura	right	no			0	18.6	16.5
6	laura	left	yes	2.1	0.8	0.724137931	14.8	14.6
7	sara	left	no	1.1	0.4	0.733333333	8.5	12.6
8	sara	right	yes	1.1	0	1	11.6	17.6
9	laura	left	yes	1	0	1	7.7	10.6
10								
11	sara	right	yes	1	0	1	10.6	11
12	sara	right	yes	2.2	0.8	0.733333333	20.5	7.8
13	laura	left	yes	0.9	0	1	6	11.3
14	laura	left	yes	1.2	0.7	0.631578947	7.5	19.8
15	sara	right	yes	1.2	0	1	9.8	17.6
16	sara	left	no	1.2	0.3	0.8	15	14.2
17	laura	left	yes	1.7	0	1	15.3	14.5
18	laura	left	yes	2	0.9	0.689655172	16.6	12
19	sara	right	yes	1.7	0	1	10.5	18.4
20	laura	left	yes	0.9	4	0.183673469	4.7	18.6
21	laura	right	no	0.9	0.9	0.5	14.1	15.8
22	sara	right	yes					
23	sara	right	yes	1.8	0	1	12	9
24	laura	right	no	2.8	0.4	0.875	9.1	18
25	sara	right	yes	1.1	0.8	0.578947368	23.4	5.2
26	sara	right	yes	0.6	0	1		

Figure 1. Uploading data into JASP.

et al. (2022). To summarize our goal in conducting this test, we want to examine the proportion of time participants spend looking at the actor who shared food with a puppet when the puppet expresses distress. Our goal in doing so is to investigate whether children interpret willingness to share saliva as indicating a particularly "thick" social relationship. Our dependent variable is the time a participant looked at the actor who shared food with the puppet divided by the time they spent looking at either actor; anything over 0.5 indicates that they looked for more than half the time at the food-sharing actor, meaning they looked longer at the food-sharing actor than at the other actor.

To use JASP for our analysis, we first need to load a file containing our data. CSV files are typically the most convenient option, but JASP accepts a variety of file types. As of this writing, most analyses in JASP require the data to be in a wide format, with a row for each participant and column headings for each variable. JASP will automatically interpret column headings as variable names, as shown along the top of Figure 1. By clicking on the column headings, we can specify each variable's type. JASP tries to do so automatically but can sometimes get confused. This step is important because JASP will restrict the analyses you can conduct to certain variable types. For example, JASP will not allow you to conduct a t-test on data from a nominal scale.

Bayesian t-test

Once we have loaded the data that we would like to subject to a t-test, we can open a Bayesian t-test module from the "T-tests" drop-down menu (*A* in Figure 2). For this example, we want the one-sample t-test.

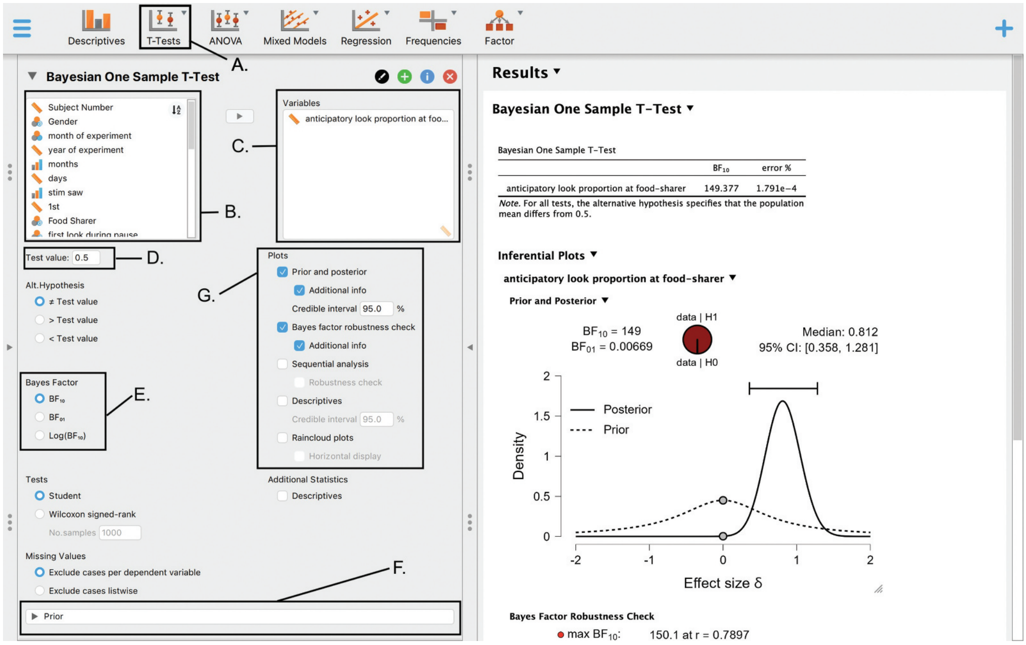


Figure 2. Main JASP menu for Bayesian one sample t-test.

Once we have opened the one-sample t-test module, the first step for running the analysis in JASP is to look at all the variables provided by our data (*B* in Figure 2), select any that we would like to analyze, and move them into the "Variables" dialog box (*C* in Figure 2) using the arrow buttons or by dragging them over. The observations are provided by the "anticipatory look proportion at food-sharer" column. As pictured in Figure 2, we move this label into the "Variables" dialog box.

Since we are conducting a one sample t-test, we need to specify the null hypothesis (*D* in Figure 2). In our current experimental context, we would set this "Test Value" to 0.5. Given our dependent variable, this value would indicate that participants are looking equally long at both actors on average.

By default, JASP performs a bidirectional hypothesis test (i.e., two-sided or two-tailed). Since this tutorial is a "quick start" guide to Bayesian methods, we retain this default approach, as is commonly done with classical t-tests.[4]

Having established that we are conducting a bidirectional t-test with a test value of 0.5, we can then choose how we want JASP to display the evidence provided by our data (*E* in Figure 2). As discussed above, we will be quantifying evidence using Bayes factors (BF). JASP defaults to BF_{01}, which will quantify our evidence in favor of the null hypothesis. We could just as easily use BF_{10}, which would quantify our evidence in favor of the alternative

[4]Our hypothesis could be considered unidirectional, suggesting a unidirectional hypothesis test: we would be *interested* if participants looked longer at either actor, but we *expect* children to look longer at the actor who shared food with the puppet when the puppet exhibits distress. Yet we find directional Bayesian t-tests to be somewhat unintuitive, particularly in regards to specifying our prior expectations. Those who are interested can refer to Dienes (2011, 2008).

hypothesis. While log (BF_{10}) is more of a niche option,[5]either BF_{01} or BF_{10} are fine to start with. We can always come back and alter our choice once we have seen our results since they convey the same information, just in different ways.

Next we come to the issue of specifying our priors (F in Figure 2). This step involves stating what we believe the population effect size might be before we conduct our experiment. As discussed in the "Specifying Priors" section above, JASP automatically uses a Cauchy distribution with a location of 0 and a scale of 0.707 as an uninformed prior. While this is an adequate default setting, JASP also provides options for specifying informed priors as needed (Figure 3).

Having supplied JASP with everything it needs to conduct a Bayesian t-test, all that remains is to select our desired outputs (G in Figure 2). It is worth noting that, while our primary focus for this tutorial is on Bayesian analysis, we should always consider and report basic descriptive statistics. We can have JASP generate a table of descriptive statistics by ticking the "Descriptives" box under "Additional Statistics."

By default nothing is selected, so JASP will only supply us with a table providing the Bayes factor.[6] While the Bayes factor is the main result from a hypothesis test, we are also typically interested in inferring the population effect size.

As discussed above, we will be quantifying our estimates of the parameter of interest using 95% credible intervals. JASP does not provide this output automatically, but we can get it by asking JASP for a plot of the prior and posterior distributions (Figure 4) and selecting the "Additional Info" option. We can choose to alter the size of our credible interval, but 95% is standard.

The plot itself visualizes how our beliefs about the parameter of interest have been updated. It also specifically includes a point representing our null hypothesis (i.e., that the population effect size is 0). The Bayes factor in favor of the null hypothesis BF_{01} is the ratio of the density at that point under the prior and posterior distribution. If our data causes that point to increase (creating a ratio greater than 1), our data has provided evidence

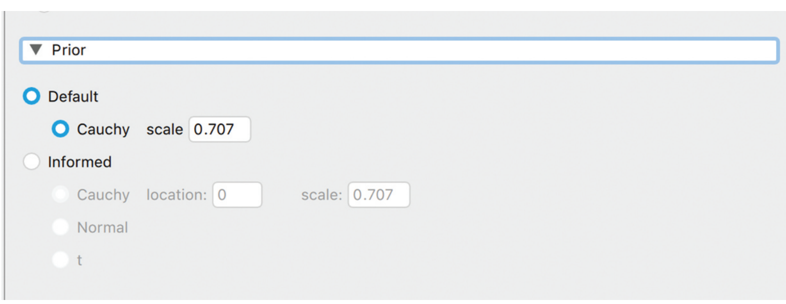


Figure 3. JASP menu for customizing t-test priors.

[5]JASP offers $log(BF_{10})$ because Bayes factors measure the weight of evidence and are thus best thought of in terms of magnitudes. As such, one of the early modern proponents of Bayesian methods proposed measuring Bayes factors in half units of log_{10} (Jeffreys, 1961).

[6]JASP also supplies an "Error %" estimate, which refers to an algorithmic shortcut used for calculating integrals. As long as the value is below 10, it is not a cause for concern and is hardly ever reported.

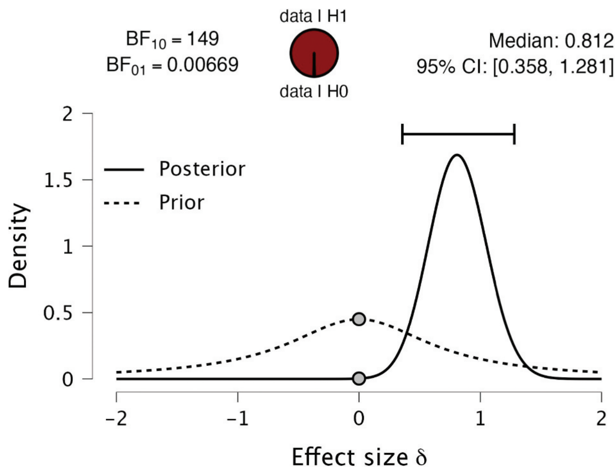


Figure 4. The "prior and posterior plot" of effect size for experimental trials.

in favor of the null hypothesis. If our data instead causes that point to decrease (creating a ratio less than 1), our data has provided evidence in favor of the alternative hypothesis.

The "Bayes factor robustness check" is also a useful option, particularly for researchers who are concerned about the influence of a specific prior. This option examines what the Bayes factor would be if we instead used particularly wide or narrow priors, indicating respectively that we are more or less willing to believe that there might be a large effect size. In other words, this option allows us to examine how big an impact our choice of priors is having on our results.

Another interesting option is "Sequential analysis," which shows us how each data point updated our beliefs as it was added to the data set. This analysis highlights a particularly useful feature of Bayesian methods, namely that we do not have to commit to a specific sample size ahead of time. Instead, it is entirely valid to continue collecting data until we have reached a predetermined threshold at which we consider our evidence to be strong enough to conclude that there is or is not an effect.

Interpreting and reporting results

In reporting the results of a Bayesian analysis, we primarily focus on the Bayes factor, which is the typical output of a Bayesian hypothesis test. We also want to consider our estimates of the parameter of interest, in this case the population effect size, so that we can better understand the nature of any effect suggested by the Bayes factor. Researchers may in fact be more interested in parameter estimation than hypothesis testing, particularly if the existence of the effect has been well-established. In any case, we also need to discuss our choice of prior, as this context is important for interpreting our results. Since a Bayesian t-test is now a routine statistical procedure, authors need only specify the prior distribution (and, for a single sample test, the test value) to give readers sufficient information to reproduce the analysis.

As discussed above, we specify our hypotheses in terms of a parameter of interest, namely the effect size. If our data increase the probability that the population effect size is 0, that means that our data are more probable under the null hypothesis: our data have provided evidence that there is no difference between the population mean and the test value. If instead our data have made it less likely that the population effect size is 0, our data are more probable under the alternative hypothesis: our data have provided evidence that there is a difference between the population mean and the test value. We find that our Bayes factor in favor of there being a difference between our population mean and the test value is approximately 150 ($BF_{10} = 149$). In other words, our data are nearly 150 times more likely under the alternative hypothesis than under the null hypothesis. Note that we can do some rounding at high values because we are most concerned with orders of magnitude. We then need to interpret this Bayes factor in terms of how *strong* the evidence is in favor of there being an effect. We might be tempted to look for thresholds at which our evidence is "sufficient" for us to reach a definitive conclusion, but such thresholds tend to be overly simplistic because we always need to consider the context of the study. In the present context, most developmental researchers would presumably agree that our data being 150× more likely under the alternative hypothesis is very strong evidence that participants do tend to look for different amounts of time at the two actors. Since readers will differ in terms of what they would consider strong evidence, we recommend a more holistic approach to evaluating results, where Bayes factors are considered in the context of the strength of past research and present methods. We return to this point in the "General Discussion."

Having established strong evidence for the existence of an effect, we can consider its direction and approximate magnitude. To do so, we typically summarize our posterior distribution for the parameter of interest by using 95% credible intervals. In this case, we see that our data have updated our beliefs about the population effect size such that we are now 95% certain that it is between 0.36 and 1.28. Based on how our dependent variable is defined, positive effects indicate that participants looked longer at the food-sharing actor.

It is also important to discuss the priors we used in obtaining our results. Since Bayes factors quantify how much our data have increased (or decreased) or belief in one hypothesis relative to another, our initial beliefs do matter. JASP's default priors are commonly used but are not universal. We should thus specify the precise nature of our priors and explain our reasons for choosing them.

In this particular instance, we might report something like the following:

> In this experiment, we tested whether children look equally long at each actor. Given our dependent variable, this behavior would be indicated by a population mean of 0.5. We thus performed a one sample t-test using JASP (JASP Team, 2023). We used a default prior for the effect size parameter, specified as Cauchy distributions with location of 0 and scale of 0.707, indicating that we generally believe smaller effect sizes are more likely than larger effect sizes. We found a Bayes factor of approximately 150 ($BF_{10} = 149$) in favor of children looking disproportionately long at one actor when the puppet exhibited distress. By examining the effect size, we can see that this difference is due to them looking longer at the actor who shared food with the puppet (posterior median = 0.81, 95% CI [0.36, 1.28]). We can interpret this result as indicating very strong evidence that children look longer on average at the food-sharing actor when the puppet exhibits distress.

Case study 2: binomial test

To provide an example of how to apply Bayesian methods to analyzing binary data, we again use data collected in the experiment discussed in the first case study (Thomas et al., 2022). To summarize, participants ($n = 27$; 16.5–18.5 months old) were shown two actors on either side of a puppet. During a familiarization phase, participants saw one actor share an orange with the puppet, thus implicitly sharing saliva, while the other actor passed a ball back and forth with the puppet. During the testing phase, the puppet exhibited distress. The researchers hypothesized that participants would interpret sharing saliva as indicating a "thicker" social relationship and would thus be more likely to look at the actor who had shared food with the puppet, expecting that actor to be the most likely to respond to the puppet's distress.

In the first case study, we analyzed data regarding how long participants looked at each actor. The researchers were also interested in which actor children looked at first, interpreting this behavior as another indication of participants' expectations regarding which actor was more likely to react.

Analysis approach

Participants had two options for which actor they could look at first, so this part of the experiment can be considered a two-alternative forced choice design (2AFC). Such designs are particularly prevalent in developmental research, where task demands can play a major role in the observed behavior of young children. Limiting participants' response options to "this or that" helps experimenters simplify experiments and reduce such concerns. In this case, our research question involves determining whether participants tend to look first at one actor more than the other.

Arguably the simplest way to analyze a 2AFC is with a binomial test. Given its simplicity, the binomial test gives us a convenient opportunity to consider the differences between the Bayesian and traditional frequentist approaches. Under the traditional frequentist approach, a binomial test would start with the premise that there is actually no tendency toward either option. In other words, we start by considering the case where our null hypothesis is correct, and thus the population-level probability of selecting the target option (i.e., the probability of a "success") is 0.5. We use the binomial distribution resulting from a probability of 0.5 to determine how likely we are to see data as extreme or more extreme than what we actually observed. For example, if we observed 8 out of 10 "successes," we would calculate the total probability of observing 8 out of 10, 9 out of 10, and 10 out of 10 successes if the actual probability of a success were 0.5. In other words, we would be measuring how likely we are to obtain data as extreme or more extreme than what we observed if the null hypothesis is correct. Indeed, this is the definition of a one-sided p-value. For a two-sided test we would also include the extremes of 0, 1, and 2 successes, which would be considered equally extreme in the other direction if the true probability is 0.5.

As we have already seen, the Bayesian approach to inference instead focuses on the relative plausibility of possible values of the parameter of interest by producing a probability distribution over those possible values. As outlined by Bayes rule, this posterior distribution is proportional to our prior beliefs about the parameter $p(\theta)$ multiplied by the probability of our data given that the parameter is a certain value $p(y|\theta)$.When the data consist of collections of binary outcomes, we can directly specify the function that defines $p(y|\theta)$ as the binomial likelihood function. The binomial distribution has a parameter dictating the probability of "success" on each trial, denoted as θ. All that remains to conduct the Bayesian binomial test is to specify a prior distribution for θ.

Specifying priors

As in the previous case study, our default approach is to use uninformed priors. Our parameter of interest θ is the population level probability of choosing one response option over another. Since θ is a probability, it can take any value between 0 and 1. As in the first case study, our first instinct might be to use a uniform distribution over all possible values of the parameter of interest. In doing so, we would be saying that we have no reason to think any particular probability value is more likely than any other value. Unlike with a t-test, where a uniform distribution over all possible parameter values is untenable, a uniform prior makes much more sense for a binomial test. Rather than an infinite parameter space with no clear boundaries, our parameter of interest for a binomial test is a probability, which is naturally bounded between 0 and 1. As such, we can simply specify our priors as a uniform distribution between 0 and 1.

Using a uniform prior presents an opportunity to highlight the relationship between our prior beliefs about a parameter ($p(\theta)$ in Box 1), the probability of the data given the value of our parameter of interest ($p(y|\theta)$), and our posterior beliefs about that parameter $p(\theta|y)$. Since we multiply $p(\theta)$ by $p(y|\theta)$ to calculate our posterior beliefs, those posterior beliefs end up being a compromise between our prior beliefs and the evidence provided by our data. Put another way, we are essentially making adjustments to what the data suggest about possible parameter values by re-weighting the evidence according to our prior beliefs. If we use a uniform prior, then our priors place equal weight on every possible parameter value, so we make no adjustments to the evidence provided by the data. This is the sense in which uninformed priors are designed to let data "speak for themselves," because we can see how our data entirely define our posterior distribution.

For the Bayesian binomial test, JASP uses the Beta distribution as the prior for θ (E in Figure 5). This distribution has two arguments that determine its shape, a and b. Beta distributions naturally stay between 0 and 1. JASP defaults to $a = 1$ and $b = 1$, which produces a uniform distribution. If we have reason to believe that some values of θ are more likely than others before we have seen our data, then we should incorporate that information into an informed prior.[7] Beta distributions where $a = b$ will be symmetrical around 0.5, indicating that we think a tendency toward the target response option is just as likely as a tendency toward the other response option. If we maintain $a = b$ but increase their value, the distribution becomes more peaked around .5, indicating that we think θ being near 0.5 is particularly likely. When a and b are unequal, the distribution skews toward one side or the other depending on whether a or b is larger. When $a > b$ we place more probability on higher values of θ, indicating that we expect a tendency toward the target response option (i.e., the participant first looks at the food sharer). A prior where $b > a$ does the opposite.

[7]It can be helpful to examine the different priors produced by altering the values of a and b by examining the "Prior and Posterior Plot" output, which we discuss below.

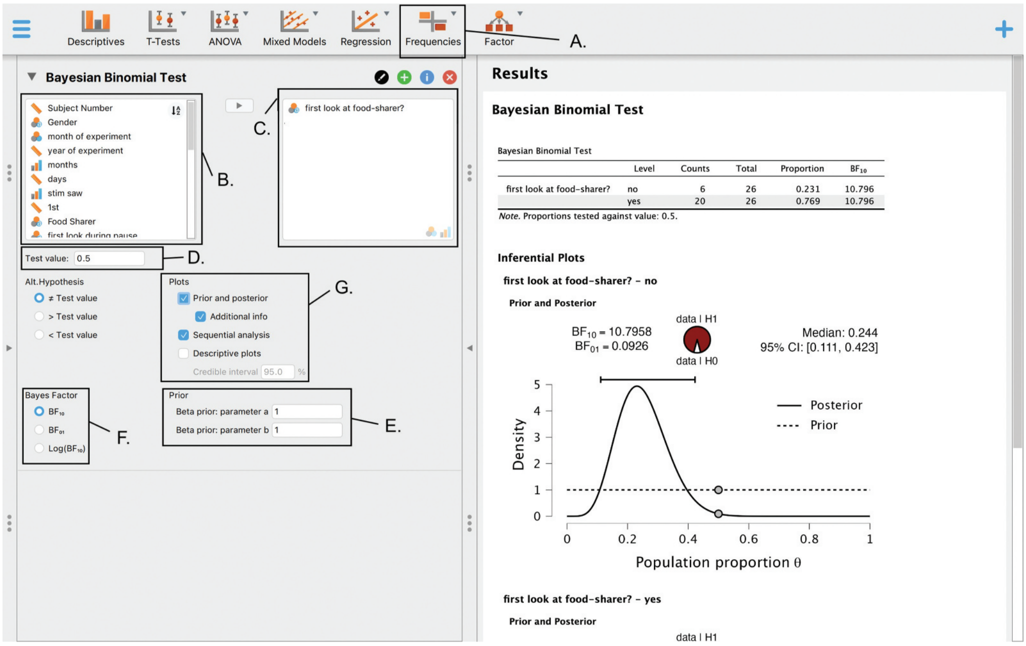


Figure 5. Main JASP menu for Bayesian binomial test.

Worked example

As always, we first need to load our data set. We again need to be careful about what JASP does in terms of assigning scale types to our data. In this case, JASP will only perform a binomial test on ordinal or nominal data.

Bayesian binomial test

Once we have loaded our data, we can select the "Bayesian Binomial Test" from the "Frequencies" module (A In Figure 5). We then need to select the column of data on which we would like to perform a binomial test. Since we are focused on who participants first looked at during the experimental manipulation, we would select the "first look at food sharer?" column from Box B in Figure 5 and then use the arrow to move it over to Box C.

The default specifications for the binomial test are already suitable for our current purposes. Our null hypothesis is that children are equally likely to look at either actor, which translates to a "Test value" of 0.5 (C in Figure 5). We could also run a directional hypothesis test, but as discussed previously, such directional tests are unintuitive and not widely used. We will also use the default uninformed priors, indicating that we think all population-level probabilities are equally plausible before seeing our data.

We can then specify the outputs we would like JASP to produce. For Bayes factors, we could have JASP quantify our evidence in favor of the null hypothesis or in favor of the alternative hypothesis. The underlying information is precisely the same in either case, but it is often best to choose whichever option produces numbers greater than 1 so that readers can think in terms of multiplication rather than fractions. As it turns out, BF_{10} produces

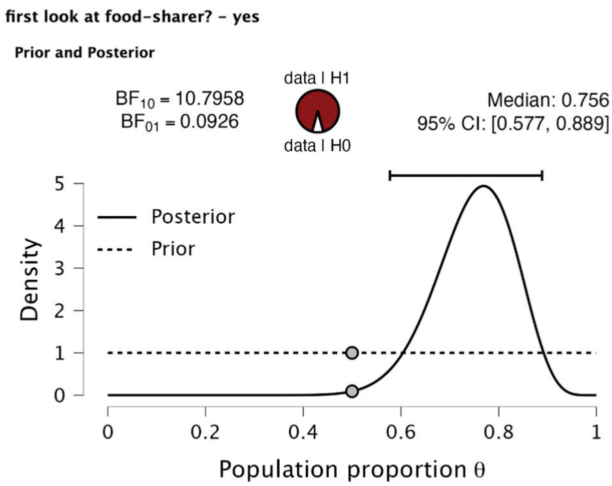


Figure 6. The "prior and posterior plot" for θ (probability of looking first at food-sharer).

a value greater than 1, indicating that our data provides evidence in favor of the alternative hypothesis, which is that children looked at one actor more than the other.

We can also ask JASP to produce various plots (G in Figure 5). The "Prior and posterior" plot (Figure 6) is particularly useful because it shows how our beliefs about the parameter of interest (θ) have been updated in light of the data. Since we designated looking first at the food-sharer as the target response option, θ represents the probability of participants looking first at the food-sharer. In this case, we can see that our prior is a uniform distribution between 0 and 1. We can also see that our data have updated our prior beliefs such that we now believe that θ is most likely greater than 0.5. Looking at the 95% credible interval similarly tells us that θ is most likely greater than 0.5.

The "Descriptive plots" do not provide additional information in this case, but the remaining option, "Sequential analysis" (Figure 7) shows us how our evidence changed as we added more data points. Under Bayesian methods, we can check our

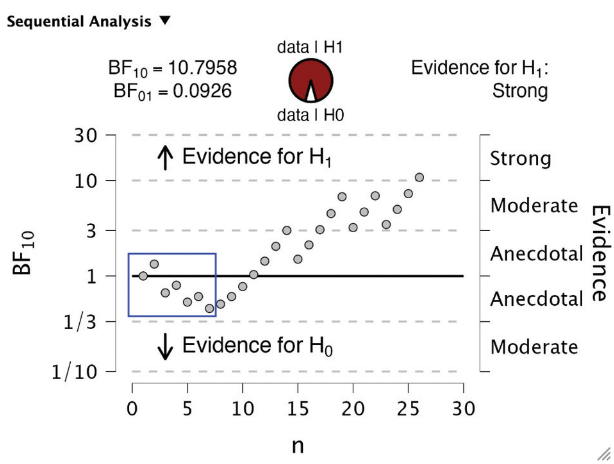


Figure 7. Sequential analysis of how BF_{10} from binomial test changes as we add each data point. Blue box emphasizes what we would infer if we stopped after 7 participants.

Bayesian Binomial Test

	Level	Counts	Total	Proportion	BF_{10}
first look at food–sharer?	no	6	26	0.231	10.796
	yes	20	26	0.769	10.796

Note. Proportions tested against value: 0.5.

Figure 8. Table summarizing results of Bayesian binomial test.

results at any point in the process, and we can stop collecting data once we are satisfied that we have strong enough evidence in one direction or the other. For example, in the analysis shown in Figure 7, we would have anecdotal evidence in favor of the null hypothesis if we had stopped after 7 participants. Intuitively, such a sample size would be far to small to draw any firm conclusions. If we listened to our intuition and kept going past 7 participants, we would indeed reach a different conclusion further down the road.

Interpreting and reporting results

Since our research question is about whether participants tend to first look at the actor who shared food with the puppet, we are primarily interested in the results of our hypothesis test. These results are summarized by the Bayes factor, which in this case is around 11 (Figure 8). A Bayes factor greater than 10 is conventionally considered strong evidence, since it indicates that our data are at least 10 times more likely under one hypothesis than a competing hypothesis.

The “Prior and posterior” plot (Figure 6) is probably our most informative figure because it shows us what we have inferred based on our data and what prior beliefs led us to that inference. The 95% credible interval summarizes this posterior distribution, showing that θ is unlikely to be smaller 0.58 and unlikely to be greater than 0.89.

To summarize these findings, we might report the following:

> To test whether participants tended to look first at the actor who shared food with the puppet, we conducted a non-directional Bayesian binomial test in JASP (JASP Team, 2022). Our null hypothesis was that the population-level probability of participants looking at the food sharer was 0.5, and our alternative hypothesis was that the probability was not equal to 0.5. Note that we coded looking at the food sharer first as a “success,” so probabilities greater than 0.5 represent a tendency to look first at the food sharer.
>
> We used an uninformed prior for θ, specified as a uniform distribution between 0 and 1. This prior indicates that we did not believe that any particular value for the probability of looking first at the food sharer was more plausible than any other probability value. We chose this prior because we wanted to make sure our data were the driving force behind our conclusions.
>
> We found that 77% of our participants looked at the food sharer first. For our hypothesis test, we obtained a Bayes factor of approximately 11 ($BF_{10} = 10.8$) in favor of participants tending to look at a specific actor, rather than being equally likely to look at either actor. When we shift our attention to parameter estimation, we can see that participants tend to look first at the food sharer (posterior median = 0.77, 95% CI [0.58, 0.89]). Taken together, we interpret these results as strong evidence that participants tend to look first at the food-sharing actor when the puppet exhibits distress.

Case study 3: correlation

For our third worked example, we will use data collected by Osborne and Suddick (1972). In this study, researchers collected data regarding students' performance on an intelligence test. More specifically, they recruited 204 children to take a version of the Weschler Intelligence Scale for Children (WISC; Weschler, 1949) before entering 1st grade and after completing 1st, 3rd and 5th grade. The test consisted of 11 subscales divided into two larger categories depending on how they were administered: performance-based or verbalized.

There are many relatively specialized statistical techniques for examining longitudinal data, including factor analysis, growth curves, and structural equation modeling. There are Bayesian versions of each of these techniques (Depaoli, 2021), but they are beyond the scope of this tutorial.

There are also many questions we might ask that can be answered with more common statistical tools. As a preliminary question, we might wonder whether a child's WISC score relative to peers is stable over time. In this data set, if relative WISC score is stable, we might expect students who perform particularly well on the WISC relative to their classmates upon entering elementary school continue to do so upon completing elementary school. In other words, to what extent is a child's performance on the WISC before 1st grade associated with their performance on the WISC at the end of 5th grade?

Analysis approach

In statistical terms, we want to describe the relationship between participants' scores before 1st grade and their scores at the end of 5th grade, both of which are continuous variables. One of the simplest ways to describe the relationship between two continuous variables is to conduct a correlational analysis.

Two of the most commonly used procedures for calculating correlation coefficients are Pearson's ρ (rho) and Kendall's τ (tau). JASP provides the Bayesian versions of each. The Bayesian version of each approach involve the same assumptions as the frequentist version (e.g., linearity, normality, homoscedasticity, continuity) because the Bayesian versions are predicated on the same underlying models as the frequentist versions. We thus need to consider which assumptions we are comfortable making when selecting our approach to Bayesian correlation. Both of our variables appear to be approximately normal, largely devoid of ties, and linearly related, so we can reasonably choose to calculate Pearson's ρ.[8] As in the preceding case studies, we need to specify our hypotheses in terms of our parameter(s) of interest to conduct a Bayesian analysis. In this case, JASP allows us to focus on the most directly applicable parameter: the population correlation coefficient (Pearson's ρ) that quantifies the strength and direction of the relationship. More extreme values represent stronger relationships, with 1 and -1 representing perfect positive and negative relationships, respectively. If the correlation coefficient is 0, there is no measurable linear relationship between the variables.

[8]The Greek letter rho signifies the population correlation coefficient, while r signifies the sample correlation coefficient. JASP refers to "Pearson's rho" because this is the parameter about which we are interested in making inferences.

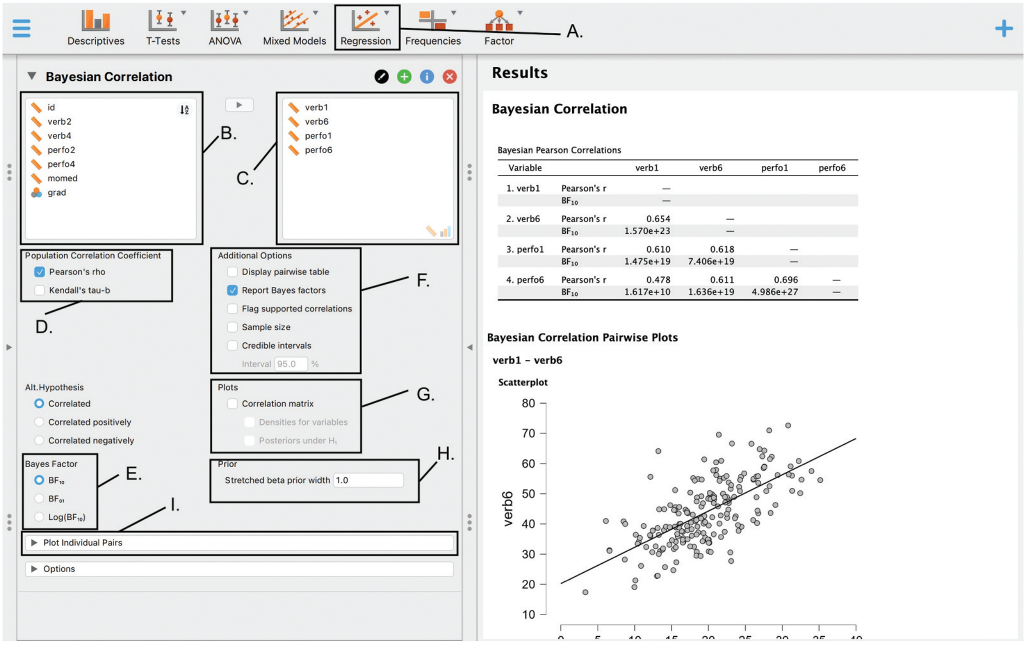


Figure 9. Main JASP menu for Bayesian correlation.

As before, we recommend defaulting to non-directional hypotheses. In other words, our default null hypothesis is that the correlation coefficient equals zero, while our alternative hypotheses is that it does not. This is JASP's default setting.

Specifying priors

Before running the analysis, we need to provide JASP with our prior beliefs regarding plausible values of the correlation coefficient (H in Figure 9). As with the binomial test, the possible values of our parameter of interest (ρ) are bounded, allowing us to specify uninformed priors as a uniform distribution between −1 and 1. This prior means that we do not think any possible value of ρ is more plausible *a priori* than any other possible value. JASP instantiates the uniform distribution as a beta distribution where $\alpha = \beta = 1$. The distribution has been stretched to range from −1 to 1.

JASP allows us to specify somewhat informed priors. Our options are limited because JASP forces $\alpha = \beta$ (creating a single parameter that it refers to as κ). This constraint means that our prior distribution for ρ must be symmetrical around 0. As a result, we cannot specify that we think a positive or negative relationship is more likely without using a directional hypothesis, which we do not recommend for most users.[9]

However, we can still specify informed priors that are symmetrical if the situation warrants it. In particular, we can specify that we think weak relationships are somewhat more likely than strong relationships or vice versa. If we think weak relationships are more likely than strong relationships, we could create a prior distribution that has a ∩ shape by

[9]There is nothing inherently problematic about specifying an asymmetrical prior; it is simply an option that JASP does not provide.

setting the parameter values to be less than 1. Since JASP parameterizes the beta distribution in terms of its width,[10] smaller parameter values create a tighter ∩ shape. The minimum value allowed by JASP is 0.003. To specify that we think strong relationships are more likely than weak relationships, we could create a prior distribution with a ∪ shape by setting the parameter values to be greater than 1. Higher values result in a more pronounced ∪ shape, indicating that we think very strong relationships are particularly likely. The maximum value allowed by JASP is 2.

Inferential output

As outputs, JASP provides information about both the value of the population correlation coefficient and about the strength of our evidence for or against our competing hypotheses. Information about the correlation coefficient comes in the form of a posterior distribution representing our beliefs about the relative plausibility of the possible values, which we can summarize with a 95% credible interval. Information about evidence for or against our hypotheses comes in the form of a Bayes factor.

These two outputs are fundamentally linked because the inferred strength of the relationship, combined with the sample size, tells us how much evidence we have for the existence of a relationship. Recall that the Bayes factor for the alternative hypothesis is the inverse of the Bayes factor for the null hypothesis, so evidence against the null is evidence in favor of the alternative and vice versa. As in the t-test example, if our data alter our beliefs such that a value of zero is less likely than we originally thought, then our data have provided evidence in favor of the alternative hypothesis. If our data make a value of zero more likely, then our data have provided evidence in favor of the null.

Worked example

As with the preceding examples, we begin by loading our data. In the process, JASP automatically identifies the type of scale in each column of data (nominal, ordinal, or scale). When planning to run a correlational analysis, it is particularly important to check that it has identified the scales correctly because JASP will not allow you to run a correlation with data it identifies as being on a nominal scale.

Bayesian correlation

Once we have loaded our data, we can select a Bayesian correlation from the "Regression" module (A in Figure 9). We then need to select the columns containing the data for the variables that we want to test for a correlation. In this case, we want to examine the relationship between participants' scores on the WISC before 1st grade and after 5th grade. As discussed above, there are two subscales, verbal and performance, so we can actually conduct a correlation analysis for each one. The relevant data for the verbal subscale are found in the "verb1" and "verb6" columns, while the data for the performance subscale are found in the "perfo1" and "perfo6" columns. Higher values indicate better performance.

[10]More specifically, the κ "width" value we supply to JASP is the inverse of the typical α and β parameters. For example, if we tell JASP that we want our priors to be represented by a beta distribution with width 0.1, it does so by stretching a beta distribution where $\alpha = \beta = 10$ across the −1 to 1 range.

Our first step is to move these four column labels from Box B into Box C using the arrow buttons (see Figure 9). We could conduct two separate analyses to examine verbal and performance scores separately, but JASP can also examine combinatorial correlations among three or more variables.

Next we need to specify the type of correlation we want to run (D in Figure 9). As discussed in the “Analysis Approach” section of this case study, we will use Pearson’s rho, but JASP makes it easy to transition to Kendall’s tau.

We then need to specify our alternative hypothesis and whether we want the evidence presented in terms of support for the null hypothesis (BF_{01}) or for the alternative hypothesis (BF_{10}) (E in Figure 9). As discussed in the “Analysis Approach” section, we suggest sticking with the default non-directional alternative hypothesis. It does not matter whether we select BF_{01} and BF_{10} at this stage. They are different formulations of the same information, so we can change our selection at any point without creating any problems. For readability, it is generally preferable to choose evidence in favor of whichever hypothesis has more support, as this prevents the reader from having to reason about fractions or decimals. JASP defaults to BF_{10} for correlations, so we will use that as our starting point.

Before worrying about what outputs we would like JASP to provide, we should also be sure to specify our priors (H in Figure 9). As discussed above in the “Priors” section of this case study, we will keep JASP’s default prior, which is a uniform distribution across all possible values of the correlation coefficient (−1 to 1).

The main output we want from JASP is the Bayes factor, which is automatically provided. A 95% credible interval is also useful for determining the direction of a correlation, assuming one exists. We can ask JASP to provide 95% credible intervals by ticking the appropriate box under “Additional Options” (F in Figure 9). We can also toggle Bayes factors on and off in the same menu, but the box is already ticked by default, and we will typically want that output.

The remaining options in this menu are not particularly impactful. We can ask JASP to display our results in a pairwise list rather than a grid by selecting “Display pairwise table.” We can also ask JASP to put asterisks next to correlations when the associated Bayes factor is more extreme than commonly referenced thresholds by selecting “Flag supported correlations,” but we will still want to consider the specific values in our specific experimental context rather than relying on arbitrary thresholds. The “Sample size” option adds a row to the default output table with the number of observations used in each correlation.

In terms of figures, the individual scatter plots for each pairing are probably the most informative, which can be selected in the “Plot Individual Pairs” menu (I in Figure 9). To use this menu, we need to tell JASP which pairs of variables we want to take a closer look at from Box B in Figure 9. The resulting plots provide descriptive information about our data points and also help readers visualize the relationship. In this case, the first pairing we are particularly interested in is “verb1” and “verb6,” so we move both from Box A to Box B in Figure 10. We are also interested in the pairing of “perfo1” and perfo6” so we also move those into the box. JASP will then automatically provide scatter plots with a line of best fit for each of the pairs we have selected. We simply need to indicate whether we are using Pearson’s rho or Kendall’s tau as our correlation coefficient.

We might also want to provide the plot illustrating how our prior distribution has been updated into the posterior distribution in light of our data (Figure 11) because it helps us visualize our remaining uncertainty regarding the population correlation coefficient.

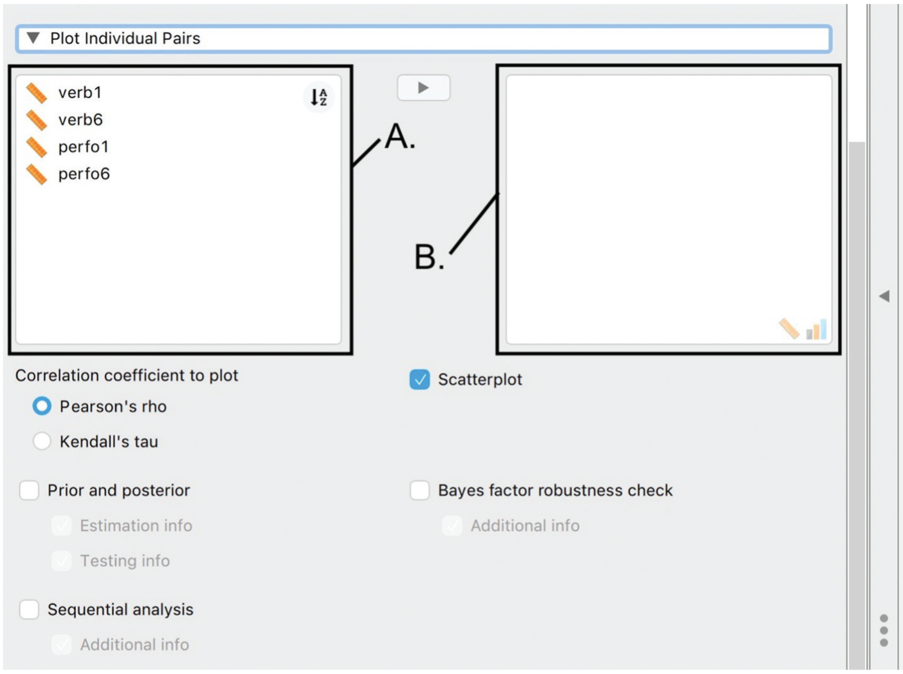


Figure 10. JASP menu for plotting individuals pairs in a Bayesian correlation.

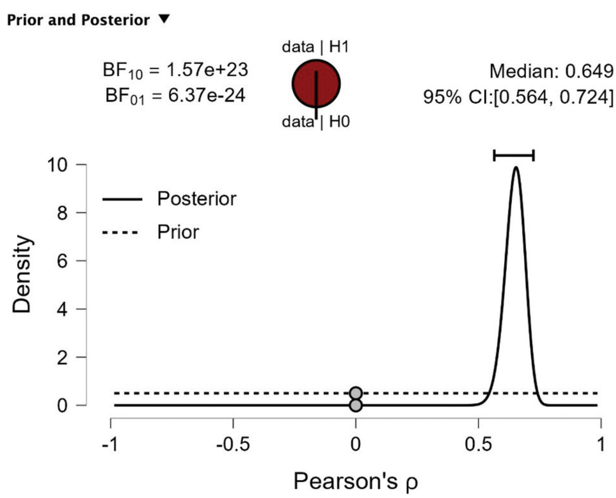


Figure 11. The "prior and posterior plot" of the correlation coefficient for verbal scores before 1st grade (verb1) and after 5th grade (verb6).

However, this information is largely captured by the 95% credible interval. The remaining plots are not typically necessary but might be helpful in certain circumstances. The "Bayes factor robustness check" plot helps reassure us that we would have reached the same conclusion regardless of what prior we chose, and the "Sequential analysis" plot shows how our evidence accumulated with each additional data point.

Bayesian Correlation

Bayesian Pearson Correlations

Variable		verb1	verb6	perfo1	perfo6
1. verb1	Pearson's r	—			
	BF_{10}	—			
2. verb6	Pearson's r	0.654	—		
	BF_{10}	1.570e+23	—		
3. perfo1	Pearson's r	0.610	0.618	—	
	BF_{10}	1.475e+19	7.406e+19	—	
4. perfo6	Pearson's r	0.478	0.611	0.696	—
	BF_{10}	1.617e+10	1.636e+19	4.986e+27	—

Figure 12. Table summarizing results of Bayesian correlation.

Interpreting and reporting results

In the default output table (Figure 12), JASP provides us with both the correlation coefficient and the Bayes factor that we have requested, quantifying the evidence regarding our competing hypotheses. In this case, we see that sample correlation is 0.654 for participants' verbal scores and 0.696 for participants' performance scores, suggesting very strong positive correlations between scores on the WISC before 1st grade and after 5th grade. We also find that $BF_{10} = 1.57 * 10^{23}$ for the verbal scores and that $BF_{10} = 4.99 * 10^{27}$ for performance scores, indicating that we have very strong evidence in favor of our alternative hypothesis that there are correlations between participants' scores before 1st and after 5th grade. The credible interval demonstrates that the correlation is almost certainly positive, since we can be 95% certain that the population correlation coefficient is between 0.564 and 0.724 for verbal scores and that it is between 0.614 and 0.758 for performance scores. In other words, relative intelligence as measured by each subscale of the WISC does seem to remain stable across elementary school.

As always, we want to report our key results as well as the priors and analysis we used in obtaining those results. In this case, we might report the following:

> We wanted to examine the stability of children's relative intelligence, as measured by the Weschler Intelligence Scale for Children (WISC; Weschler, 1949), across elementary school. In other words, to what extent is their score on the test before entering 1st grade associated with their score on the test after completing 5th grade? If children develop at similar rates across elementary school, we would expect children whose scores are particularly high or low before 1st grade to still have particularly high or low scores upon completing 5th grade.
>
> To examine this question, we used data collected by Osborne and Suddick (1972), in which 204 children took the WISC at multiple points throughout elementary school. We analyzed children's scores on the WISC's verbal and performance subscales individually, looking for a correlation between their performance before entering 1st grade and their performance after completing 5th grade. Since children's performance on both subscales was normally distributed at both time points, and since the relationship between the two is fairly linear, we performed a Pearson's rho correlation using JASP (JASP Team, 2023). We used the default uninformed priors for possible values of rho, namely a uniform distribution between −1 and 1. This prior indicates that we do not consider any possible correlation coefficient value as more likely than any other before examining the data.

We found strong evidence in favor of a positive correlation between children's scores before entering 1st grade and after completing 5th grade. For the verbal subscale, we estimate a positive correlation coefficient of 0.654 (95% CI [0.564, 0.724]). For the performance subscale, we estimate a positive correlation of 0.696 (95%CI [0.614, 0.758]). In line with these estimations, we find Bayes factors that indicate we have very strong evidence for both correlations:

$BF_{10} = 1.57 * 10^{23}$ and $4.99 * 10^{27}$ for the verbal and performance subscales, respectively. We interpret these results as indicating that children's relative in- telligence, as defined by the WISC, is quite stable across their time in elementary school.

Case study 4: linear regression

In the preceding case study, we established that there is a strong correlation between children's performance on the Weschler Intelligence Scale for Children (WISC; Weschler, 1949) before 1st grade and after 5th grade. We reached this conclusion by analyzing a data set collected by Osborne and Suddick (1972). This result suggests that children's relative intelligence, at least as measured by this test, is stable across elementary school.

However, given the number of factors that can influence students' scores on this type of test, we might suspect that something else may be driving the association between WISC scores before 1st grade and after 5th grade. In particular, aspects of socioeconomic status, such as parents' education, can be an important predictor of performance on intelligence tests (Rowe et al., 1999). In collecting the data we have been using, Osborne and Suddick also recorded the educational attainment of participants' mothers in terms of years of schooling. In this case study, we will examine whether children's scores on the WISC before 1st grade explain their scores after 5th grade once we control for their mother's educational attainment.

Analysis approach

Since our research question involves considering the relationship between two variables after adjusting for a third variable, we would naturally use multiple linear regression. Given the likely causal relationship between these three variables, we will treat score on the WISC after 5th grade as the outcome variable, with score on the WISC in 1st grade and mother's education as possible predictors of that outcome.

Bayesian linear regression is based on comparisons between linear models that contain certain combinations of the predictors. Linear models that include more predictors can account for a greater variety of data patterns, meaning that they will never have *more* prediction error than less complex models. Model comparison techniques therefore focus on whether the extra complexity created by incorporating variables as predictors reduces error enough to be "worth it." When the additional model complexity results in a markedly better fit to the data, there is evidently a relationship between the outcome and the predictors. The specific models to be compared will depend on the research question.

In the present example, there are two potential predictors. If we consider all of the models we could make with these two predictors, the most complex incorporates both (the full model), while the simplest incorporates no predictors (the null model). Intermediate models would only incorporate one predictor or the other.

Since we want to know whether the previously discovered correlation between scores before 1st grade and after 5th grade is confounded by mother's education, only certain comparisons are meaningful. For example, we are not particularly interested in the null model because we already know that it will never be the best performing model. If the model that only includes the 1st grade scores as a predictor is the best performing model, then that suggests the correlation between 1st and 5th grade scores is not confounded by mother's education. If the model that only includes mother's education as a predictor is the best performing model, then that suggests the correlation between 1st and 5th grade scores is wholly confounded by mother's education. If a model with both predictors outperforms the models that include only one of them, we have evidence that both predictors are related to 5th grade scores. Such a result could indicate that scores after 1st grade are only partially confounded by mother's education, or that mother's education has an independent association in addition to that of 1st grade scores. In any case, the results of the previous case study make it inevitable that the null model will be inferior to models that include one or both of the predictors.

Should we find evidence that both scores before 1st grade and mother's education relate to scores after 5th grade, then we need to investigate parameter estimates to determine the roles of the two predictors. More specifically, we would want to examine how the parameter estimates associated with scores after 1st grade change as we add and remove the additional mother's education predictor. Each predictor in a model has a coefficient that represents the effect of that predictor. In this example, how much the the apparent relationship between 1st grade scores and 5th grade scores is confounded by mother's education corresponds to how much the coefficient associated with 1st grade decreases when we switch from the model where that is the only predictor to the full model.

As in any situation where we are estimating parameters that are used in multiple competing models, we need to be mindful of whether we are considering our general uncertainty about the parameter estimates across all possible models or our specific uncertainty about a parameter estimate in the context of a specific model. Parameter estimates depend on which model we are considering, and the relative performance of a model depends on parameter estimates. By default, JASP provides us with information about parameter estimates averaged across all models by weighting the parameter estimates produced by each model in proportion to how likely each model is, a procedure known as model averaging (Hinne et al., 2020). This procedure works well in many situations, as it explicitly accounts for multiple forms of uncertainty. If we are uncertain regarding what models we should consider and still need to make decisions based on parameter estimates in spite of that uncertainty, we might focus on model averages rather than on specific models. Yet in the context of our current question, doing so would involve incorporating models that we have already discounted into that average. Instead, we focus on parameter estimates from specific models of interest. This procedure requires some tinkering, which we demonstrate in the worked example below.

For simplicity, we will only consider additive effects here, rather than interactions. This approach makes sense in the context of our research question, which is to examine whether the influence of scores before 1st grade on scores after 5th grade is confounded by mother's education. However, we might suspect that the effect of 1st grade scores varies depending on mother's education. For example, we could reasonably suspect that mother's education is more impactful for students who had low scores in 1st grade compared to those with higher scores. To examine such a possibility, we would simply add a third predictor accounting for the interaction. With three predictors (1st grade scores, mother's education, and an interaction term) 8 models could be constructed (null, full, and 6 intermediate models), although not all of those models may be appropriate in a given situation. We mention the steps for doing so in JASP in the "Worked Example" section below.

Specifying priors

As in any Bayesian analysis, we must set our prior expectations appropriately. In this case, we have two types of prior expectations: our expectations regarding which model best explains the data and our expectations for the parameters in each model.

For our expectations regarding which model best explains the data, we might try to make as few commitments as possible by specifying that we do not believe any given model to be more likely than any other model before we have seen our data. This belief would correspond to a uniform prior over all possible models. In cases with a small number of predictors, this may be sensible. However, if we have many predictors, there will be a null model, a full model, and numerous intermediate models made up of various combinations of predictors. Inevitably, there will only be a single null model and a single full model, with the rest of the models being intermediate. Thus, if we assign equal probability to all models, we end up assigning most of our probability to intermediate models. In other words, by committing to a uniform prior over all possible models, we are implicitly saying that we will very likely conclude that some but not all predictors matter.

To avoid committing to such a belief, we could instead specify our priors in terms of how many predictors we think the model should have. Indeed, JASP defaults to such a prior, using a beta-binomial distribution to create a uniform distribution over number of predictors. In other words, we divide the probability evenly between categories of models: those that have 0 predictors, those that have 1 predictor, etc. We then take the probability assigned to each category and divide it evenly between the specific models within that category. As a consequence, the null model and the full model have equal probability because they are each the only models in their category. Each intermediate model has less probability because there are more versions of models with that many predictors.

The beta-binomial prior only substantially differs from the uniform prior when we are considering many possible predictors. If we are only considering one predictor, then we just have the null model and the full model, each with equal probability. In such a situation, the beta-binomial prior and uniform prior are equivalent. When we have 2 predictors, the beta-binomial prior specifies that we think the null and full models both have a probability of 0.333, while the 2 intermediate models have a combined probability of 0.333, meaning that each of them has a probability of 0.167. This prior is not all that different from a uniform prior, which would assign each of the four possible models a prior probability of 0.25. As we continue to add predictors, though, the difference becomes more pronounced. If the choice between a beta-binomial prior and a uniform prior is a cause for concern, a single click in JASP allows us to change our priors and see what effect the choice has on our conclusions. And when the data provide strong evidence, the nuances of our specific choice of priors will not materially alter our conclusions.

For the parameter(s) within each model, our priors are similar to those we use for a t-test: a multivariate Cauchy distribution with a location of 0, a scale of 0.354, and as many

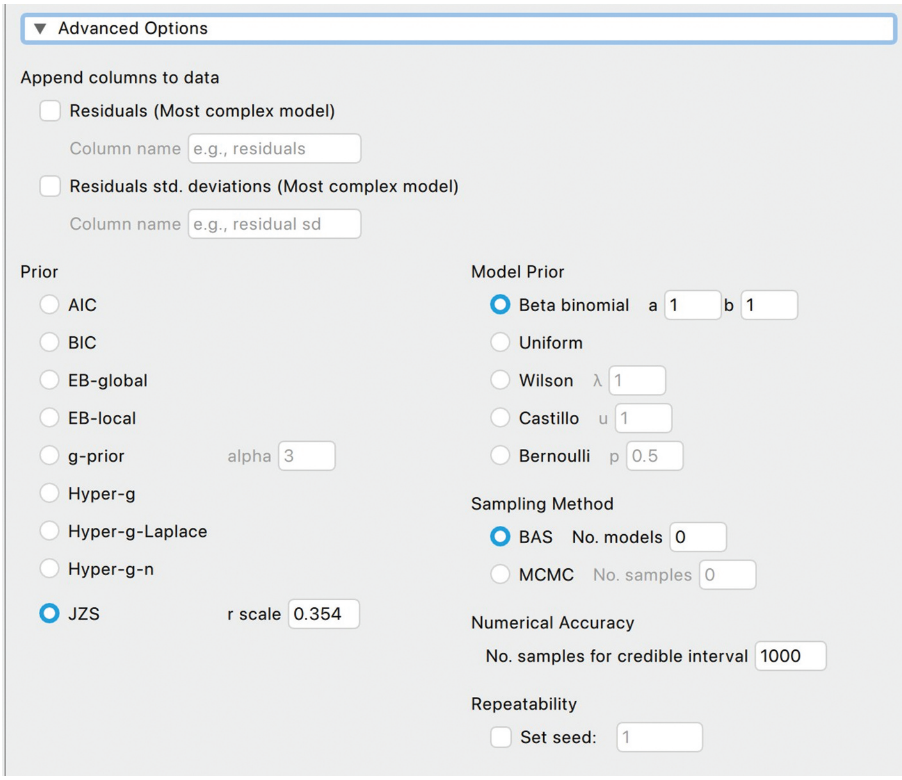


Figure 13. JASP menu for customizing linear regression priors.

dimensions as we have parameters. We want to make as few commitments as possible, so we simply say that we think smaller standardized effects are more likely than larger standardized effects. As with the default prior for the t-test, locating the Cauchy distribution at 0 means that we are agnostic about the direction of any effects. In the case of a linear regression, our effects are defined by the slope associated with each predictor. We standardize the slopes by estimating our line of best fit on data that has itself been standardized, which involves subtracting the sample mean from each data point and dividing the result by the sample standard deviation. Increasing or decreasing the scale of the Cauchy distribution indicates that we believe large effects are respectively more or less likely. JASP allows for further customization by offering a wide array of possible distributions (under "Advanced Options;" Figure 13), but these alternative priors are beyond the scope of the present paper.

Inferential output

As with the preceding case studies, our two main inferential outputs are Bayes factors and 95% credible intervals. Bayes factors allow us to weigh the evidence for two competing hypotheses. Credible intervals allow us to summarize our updated beliefs about the likely size and direction of effects, assuming they exist.

With a Bayesian linear regression, both of these inferential outputs become somewhat more complicated than they are for a t-test, a binomial test, or a correlation. The credible intervals are more complex because each predictor can have its own effect, as can interactions between the predictors. This situation simply means that we have multiple credible intervals to consider, since we have multiple parameters of interest. As discussed in "Analysis Approach," another complicating factor is that our beliefs about a predictor depend on what other predictors are included in a model. In other words, we need to be mindful that our parameter estimates depend on the models in which those parameters are situated.

Bayes factors become more complicated in Bayesian linear regression because Bayes factors are used for comparing two competing hypotheses, and with a linear regression, we are often considering more than two hypotheses. In the present example, we are considering the possibility that the relationship between scores before 1st grade and after 5th grade is (1) not at all confounded, (2) fully confounded, and (3) partially confounded by mother's education. We test these hypotheses by comparing pairs of models. These pairwise model comparisons produce Bayes factors quantifying the evidence in favor of a given reference model performing better than a given competing model (BF_{01}), or in favor of the competing model performing better than the reference model (BF_{10}). We can then interpret the results of these model comparisons in terms of what they imply about our hypotheses.

In the spirit of being thorough, our first instinct might be to compare each alternative model to the null model. Such an approach would indeed allow us to test all of our hypotheses, since we can use the common reference point of the null model to determine how the alternative models compare to each other. For example, if one model comparison produces a Bayes factor indicating that the data is 2× more likely under Model A than under the null model ($BF_{10} = 2$ in favor of Model A), and a second model comparison indicates that the data is 8× more likely under Model B than under the null model ($BF_{10} = 8$ in favor of Model B), we can conclude that the data is 4× more likely under Model B than under Model A. However, the null model is not typically the most interesting point of comparison. In the present example, we have in fact already determined that the null model is irrelevant, because our previous worked example showed that scores before 1st grade and after 5th grade are correlated.

For a more efficient option, we can instead use the best performing model as our reference point, which JASP does by default. This default means that $BF_{10} = 1$ for the best model, since it is being compared to itself. The question is how much *worse* alternative models do in comparison, so all remaining BF_{10} values will be less than 1. Continuing with our example from the previous paragraph, Model B is the best model and thus acts as the common reference. When we compare Model B to the reference model (i.e., itself), we find that $BF_{10} = 1$. Model B was 4× better than Model A, so we would find that $BF_{10} = .25$ in favor of Model A. Finally, Model B was 8× better than the null model, so we would find that $BF_{10} = .125$ in favor of the null model. All of these options are just different ways of presenting the same information, but for simplicity, we recommend sticking with the defaults.

Alternatively, JASP provides Bayes factors that summarize the evidence in favor of including each individual predictor ($BF_{inclusion}$). These Bayes factors are essentially a reformulation of the information provided by the Bayes factors comparing specific models. To calculate them, JASP compares the performance of models that contain the predictor to those that do not. Yet in doing so, JASP considers models that we have already discounted in the present example. Thus we will instead focus on specific model comparisons.

Beyond Bayes factors, JASP also provides information about parameter estimates. As with other such outputs, this information comes in the form of descriptive summaries of the marginal posterior distribution for each parameter.[11] These descriptives again include the center, spread, and 95% credible interval of the posterior distribution.

Worked example

Once again, we first load our data. JASP automatically labels each column of data according to the type of scale on which that data is measured, but it is worth checking to make sure that JASP has done so correctly.

Bayesian multiple linear regression

Having loaded our data, we can select the “Bayesian Linear Regression” option from the “Regression” module along the top of the screen (A in Figure 14). We then need to identify the data that will be our dependent variable and the data that will be our predictor(s) (covariates) from Box B and then place them in the appropriate boxes, either by dragging the label or by highlighting the label and using the arrow buttons.[12] In this case, we consider performance on the WISC after completing 5th grade to be our dependent variable, and we are interested in how a student’s performance on the WISC before 1st grade and their mother’s education are associated with that dependent variable.

As in the previous case study, we can examine students’ performance on both verbal and performance-based sections of the WISC. We can only examine a single dependent variable at a time when conducting a linear regression, so we need to analyze each section of the WISC in turn. We could first look at verbal performance by moving “verb6” from Box A into Box B using the appropriate arrow button (see Figure 14), and we would move “verb1” and “momed” into Box C. To subsequently examine another dependent variable (i.e., “perfo6”), we could once again select the “Bayesian Linear Regression” option from the “Regression” module. Doing so will create an additional analysis menu below the original analysis menu in the left half of the screen, and the new outputs will be placed in an additional section below the original outputs on the right half of the screen.

We then need to specify whether we want to quantify our model comparison results in terms of evidence in favor of (BF_{10}) or against (BF_{01}) a given model (C in Figure 14). We also need to specify whether we want to use the null model or whichever model is best as our reference point for the comparisons (D in Figure 14). As discussed in the “Inferential Output” section above, we recommend keeping the default settings (“BF_{10}” and “Compare

[11]The full posterior distribution is multidimensional, accounting for every combination of parameter values. JASP summarizes this multidimensional posterior using all of its marginal posterior distributions for each given parameter. For example, for three parameters we would have the joint posterior $p(\theta_1, \theta_2, \theta_3|y)$. A marginal distribution, say $p(\theta_1|y)$, is obtained by integrating out θ_2 and θ_3.

[12]If we want to examine an interaction, we would need to use the “Model” drop down menu. The dialogue box on the right shows the terms that are currently being considered in various models, while the dialogue box on the left shows available predictors, which is based on the columns of data selected in “Covariates” box from the original menu. To incorporate an interaction term, we need to select multiple predictors in the left dialogue box by clicking one predictor and then holding the “Ctrl” or “Shift” key when clicking on more. Pressing the arrow while multiple predictors are highlighted, or simultaneously dragging multiple highlighted predictors to the right dialogue box, will prompt JASP to create an interaction term. For example, selecting and simultaneously dragging both perfo1 and momed to the right dialogue box will prompt JASP to add a “perfo1*momed” interaction term.

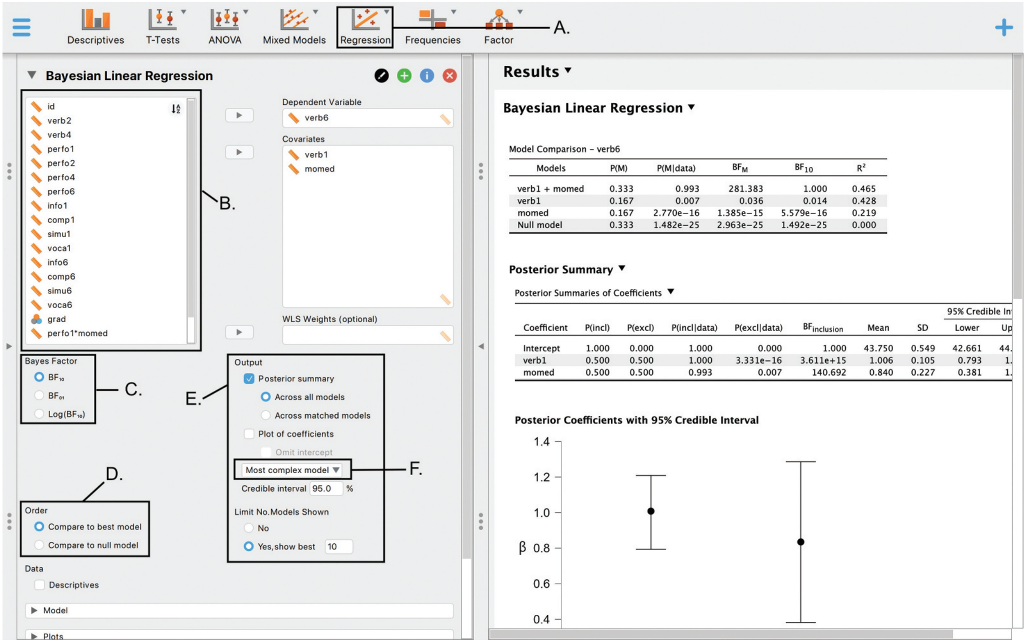


Figure 14. Main JASP menu for Bayesian linear regression.

to best model"). We will also be using the default uninformed priors, as discussed above, but priors can be customized in the "Advanced Options" menu (Figure 13).

For outputs, JASP defaults to providing only a table with 6 columns summarizing the model comparison results, but we are primarily interested in the fifth column (Figure 15). The first column lists the predictors in each model. The second column, "$P(M)$," provides the prior probability of each model. As discussed in the above "Specifying Priors" section, we use JASP's default of a beta binomial prior rather than a uniform prior, so some of the models differ in their prior probability. The third column, "$P(M|\text{data})$," provides the posterior probability of each model, accounting for both our priors and the data. The fourth column, "BF_M," quantifies how our beliefs about the plausibility of the various models has changed in light of our data (i.e., from prior to posterior). This fifth column is the most interesting to us because it provides Bayes factors that compare the performance of the various models, so we will discuss it in greater depth when we interpret and report our results. The sixth column, "R^2," quantifies the proportion of the total variance in the dependent variable that can be accounted for by each model.[13]

Beyond these default outputs, JASP provides additional options (E in Figure 14). The "Posterior summary" provides details regarding our inferences about each of the coefficients for each of the predictors being mixed and matched in the various models we are comparing (Figure 16). The distinction between evaluating coefficients "Across all models"

[13]Note that models with more predictors will always account for at least as much variance as models with fewer predictors, simply because they are inherently more flexible. As such, R^2 is not the best means of comparing models because we need to consider whether a model's extra complexity can account for enough variance to be justified. Instead, we focus on BF_{10}, which balances both fit and complexity.

Model Comparison – verb6

Models	P(M)	P(M|data)	BF_M	BF_{10}	R^2
verb1 + momed	0.333	0.993	281.383	1.000	0.465
verb1	0.167	0.007	0.036	0.014	0.428
momed	0.167	2.770e−16	1.385e−15	5.579e−16	0.219
Null model	0.333	1.482e−25	2.963e−25	1.492e−25	0.000

Model Comparison – perfo6

Models	P(M)	P(M|data)	BF_M	BF_{10}	R^2
perfo1	0.167	0.805	20.596	1.000	0.484
perfo1 + momed	0.333	0.195	0.486	0.121	0.486
momed	0.167	6.411e−23	3.205e−22	7.967e−23	0.137
Null model	0.333	5.564e−28	1.113e−27	3.458e−28	0.000

Figure 15. Tables summarizing model comparison results for Bayesian linear regression.

or "Across matched models" only applies to situations in which an interaction is being considered.[14]

When it comes to interpreting the posterior summary, the "*P* (incl)," "*P* (excl)," "*P* (incl|data)," "*P* (excl|data)," and "$BF_{\text{inclusion}}$" are columns we will not be focusing on in our example (for details about these see van Doorn et al., 2021; Wagenmakers et al., 2018). The remaining columns provide information about the specific values of the parameters. When using Bayesian methods, we quantify our beliefs about the values of parameters of interest with a probability distribution. Thus JASP provides descriptive information about those probability distributions, including the central tendency ("Mean"), variation ("SD"), and "95% Credible Interval."

To make sure that the posterior summary provides the information we want, we need to indicate how we would like JASP to calculate it. JASP defaults to model-averaging, but as discussed above, this approach considers models that we have already discounted. Instead, we would like to base our parameter inferences on specific models. We can get parameter inferences specific to each possible model by selecting "Most Complex Model" in Box F of Figure 14 and then moving potential predictors in and out of the "covariates" box. By selecting "Most Complex Model," we are telling JASP to provide us with parameter inferences from the model with the most predictors. We can manipulate what the most complex model will be based on what predictors we allow JASP to consider. In our case, we are particularly interested in comparing the model that includes both mother's education and 1st grade scores as potential predictors to a model that only considers 1st grade scores. For example, if we include both "verb1" and "momed" in the "covariates" box, the most complex model will include both of those predictors. If we remove "momed" from the "covariates" box, the most complex model will only include "verb1" as a predictor. We can then examine how our inferences for "verb1" change based on whether "momed" is also

[14]This option addresses a debate over whether a model should be considered if it includes an interaction term but does not include the constituent main effects (van den Bergh et al., 2020).

Posterior Summary

Posterior Summaries of Coefficients

								95% Credible Interval	
Coefficient	P(incl)	P(excl)	P(incl|data)	P(excl|data)	$BF_{inclusion}$	Mean	SD	Lower	Upper
Intercept	1.000	0.000	1.000	0.000	1.000	43.750	0.549	42.661	44.802
verb1	0.500	0.500	1.000	3.331e−16	3.611e+15	1.006	0.105	0.793	1.208
momed	0.500	0.500	0.993	0.007	140.692	0.840	0.227	0.381	1.285

Posterior Summary

Posterior Summaries of Coefficients

								95% Credible Interval	
Coefficient	P(incl)	P(excl)	P(incl|data)	P(excl|data)	$BF_{inclusion}$	Mean	SD	Lower	Upper
Intercept	1.000	0.000	1.000	0.000	1.000	50.932	0.629	49.541	52.024
perfo1	0.500	0.500	1.000	3.458×10^{-28}	$2.892\times10^{+27}$	1.030	0.075	0.864	1.161

Figure 16. Table summarizing model-specific parameter estimates for the best performing models in each Bayesian linear regression. Note that we only have estimates for "perfo1" in the second table because model comparisons indicate that the best performing model does not include mother's education as a predictor.

included in the model. The greater the change, the more confounding there is from mother's education.

JASP also provides various options for plots, both in the initial menu and under the "Plots" tab. Setting aside plots that help check the assumptions involved in any linear regression, we are typically most interested in plots of the posterior distributions for each parameter of interest. One such plot is the "Plot of coefficients" available in the initial menu, which includes a point for the mean of each posterior distribution and error bars representing the 95% credible intervals. It is often best to omit the intercept from these plots because the intercept may be uninterpretable. Furthermore, the intercept makes an additive contribution to the dependent variable, while predictors make a multiplicative contribution, so including the intercept may distort the scale of the plot. Under the "Plots" tab, there is also the option to plot the "Marginal posterior distributions," which will visualize the entire posterior distribution for each parameter, rather than just the point and interval summaries. But since each distribution is plotted individually, these plots are not as conducive to direct comparisons.

Interpreting and reporting results

In this case, we see different results depending on whether we are looking at scores from the verbal or performance sections of the WISC. For the verbal scores, we can see that our best performing model includes both scores before entering 1st grade and mother's education, so we have evidence that both scores before 1st grade and mother's education predict scores after 5th grade. To quantify this evidence, we can look at the BF_{10} comparing the full model to the reduced model. The BF_{10} for the model including "verb1" and "momed" is 1 because it is the best perform- ing. The model that only includes "verb1" has a BF_{10} of 0.014. We can take the inverse of BF_{10} to find a Bayes factor of 1/0.014 = 71.4 in favor of *including* mother's education in a model of scores on the verbal section after 5th grade, which is typically considered strong evidence (Jeffreys, 1961).

In contrast, we can see that our best performing model for performance scores only includes scores before entering first grade. Mother's education is evidently not an important predictor of scores on the performance section of the WISC after 5th grade, so have evidence against the relationship between scores on the performance section being confounded by mother's education. We can once again inspect the Bayes factors to quantify this evidence. The BF_{10} for the model including only "perfo1" is 1 because it is the best performing. The model that includes both "perfo1" and "momed" has a BF_{10} of 0.121. We can again take the inverse of BF_{10} to find a Bayes factor of $1/.121 = 8.26$ in favor of *excluding* mother's education from a model of scores on the performance section after 5th grade, which might be considered either weak or moderate evidence depending on the context.

We now turn to the parameter estimates to examine the extent to which mother's education confounds the relationship between scores before 1st grade and after 5th grade on the verbal portion of the WISC. We apply the procedure described in the worked example to the verbal data only, because the model comparison results suggest that the relationship between the performance scores is not confounded. Our estimates for "verb1" do change somewhat when we include "momed" as a predictor. In a model that only includes verb1 as a predictor, we find a mean estimate of 1.187 (95% CI [0.996, 1.339]). When we add in "momed," that decreases to 1.006 (95% CI [0.798, 1.214]), an approximately 15% reduction. In addition, our parameter estimate for "momed" has a mean of 0.840 (95% CI [0.393, 1.288]). Thus we can conclude that mother's education does somewhat confound the relationship between scores before 1st grade and after 5th grade, but that both 1st grade scores and mother's education are strong predictors in their own right.

In this instance, we might report the following:

> We already have strong evidence of a correlation between children's scores on the Weschler Intelligence Scale for Children (WISC; Weschler, 1949) before entering 1st grade and after completing 5th grade. Next, we examine whether this relationship might be confounded by socioeconomic status. To do so, we continued using the data set collected by Osborne and Suddick (1972). In addition to recording participants' scores on the WISC, the data also includes information about the education of participants' mothers (measured in years), which we used as our operational definition of socioeconomic status.
>
> We analyzed scores on the verbal and performance subscales of the WISC separately. For each subscale, we conducted a Bayesian linear regression analysis in JASP (JASP Team, 2023), with mother's education and scores before 1st grade as potential predictors of scores after 5th grade. More specifically, we compared 4 possible linear models: a null model, a model with only 1st grade scores as a predictor, a model with only mother's education as a predictor, and a full model with both predictors. For simplicity, we did not consider potential interactions between mother's education and 1st grade scores. We used the default uninformed priors for both model comparison and for estimating coefficients within each model. For model comparison, the default uninformed prior is a beta-binomial distribution, which indicates that we believe models with two predictors (full model), one predictor, or no predictors (null model) are equally likely. For coefficients, the default uninformed prior is a multivariate Cauchy with a location of 0 and a scale of 0.354, indicating that we generally believe small effects are more likely than large effects.
>
> In examining scores on the verbal sections of the WISC, we found evidence suggesting that mother's education partially confounds the relationship between scores before 1st grade and after 5th grade. Specifically, we found a Bayes factor of approximately 71.4 in favor of including

mother's education as a predictor (in addition to scores before 1st grade, as indicated by the previously identified correlation). Closer examination of the posterior distributions reveals that the estimated influence of 1st grade scores on 5th grade scores was 1.187 (95% CI [0.996, 1.339]) when it is the only predictor included in the model. This value decreases to 1.006 (95% CI [0.798, 1.214]) when mother's education is included as an additional predictor. This 15% reduction in the estimate indicates partial confounding; the scores before 1st grade and mother's education account for the some of the same variation in scores after 5th grade. However, 1st grade scores and mother's education are each strong predictors of scores after 5th grade in their own right.

In contrast, we found moderate evidence that mother's education does not confound the relationship between scores on the performance sections of the WISC before 1st grade and after 5th grade. Specifically, we found a Bayes factor of approximately 8.3 in favor of only including 1st grade scores (and excluding mother's education) as a predictor of scores after 5th grade.

Case study 5: ANOVA

For our final case study, we will use data collected by Clegg and Legare (2016). Past research in Western societies (Legare et al., 2015) has indicated that children tend to be less committed to high fidelity imitation when they are being shown how to accomplish a goal (instrumental behavior) than when they are shown how to perform rituals (conventional behavior). Clegg and Legare subsequently collected cross-cultural data to examine whether children's fidelity in imitating adults may differ between Western and non-Western cultural settings.

Participants (142 6–8-year-olds) were recruited from classrooms in the US or on the island of Tanna in Vanuatu. Participants were then assigned to observe an adult performing either an instrumental or conventional behavior. All participants observed a researcher demonstrating how to make a necklace. Participants assigned to the instrumental condition were explicitly told that the goal of the demonstration was to show them how to make a necklace. Participants assigned to the conventional condition were told that "people always do it like this," implying that the goal of the demonstration was for them to learn a traditional way of making a necklace. Participants were then asked to make a necklace themselves. Their fidelity to the original demonstration was measured in terms of how many of 5 target behaviors they copied, resulting in a "fidelity score" from 0 to 5.

We can summarize this design as a 2×2 between-subjects experiment: we have a factor of country (US vs. Vanuatu) and a factor of condition (instrumental vs. conventional). We are interested in whether the fidelity with which participants imitate a demonstration varies across countries, across conditions, or both.

Analysis approach

Traditionally, developmental psychologists have analyzed such data using Analysis of Variance (ANOVA). More recently, researchers have begun to favor more general regression models (see Muradoglu et al., 2023). However, ANOVAs are so ubiquitous in the literature that we believe it is worth discussing how to run them from a Bayesian perspective. Furthermore, ANOVAs are fundamentally a specific instantiation of a regression model, and thus the process of interpreting the results of a Bayesian ANOVA is similar to how one interprets the results of a Bayesian regression.

As pointed out by Rouder et al. (2016), there are two ways of approaching ANOVAs. The one most commonly taught in introductory statistics courses is the procedural approach, which involves calculating an F statistic that compares the variation within factor levels to the variation between factor levels. The Bayesian equivalent of ANOVAs instead uses a model-comparison approach, similar to regression, which involves comparing linear models that either include or ignore factor-level distinctions as predictors.

In both the procedural and model-based approaches to ANOVAs, we are essentially treating our factor levels as a system for categorizing results and then testing whether that categorization system is good (i.e., helps us better predict results than ignoring those categories). If we have more than two levels for a factor, we might want to use post-hoc tests to see which levels are useful categories (i.e., meaningfully differ from each other). However, if we have only two levels for a given factor, information about whether we should include that distinction in our categorization system inevitably tells us that there is a meaningful difference between those two levels, making a post-hoc test unnecessary.

To fully test the categorization system in our present example, we compare 5 models: (1) the null model, (2) a model that only accounts for the location in which participants were recruited, (3) a model that only accounts for the condition to which participants were assigned, (4) a model that accounts for both location and condition independently, and (5) a full model that accounts for both location and condition as well as an interaction between the two.[15] While we can use post-hoc tests if certain pairwise comparisons are of particular interest, they are unnecessary in the present example. Since each factor only has 2 levels, we can use the model comparisons to fully investigate each factor and interaction, looking at how including a given factor or interaction impacts predictions to determine whether each factor has an effect.

In the present example, our primary focus is on whether effects exist. Is there a difference between US responses and Vanuatu responses, regardless of condition? Is there a difference between responses in the instrumental condition and the conventional condition, regardless of country? Are the differences between conditions consistent across countries? We answer such questions by comparing models that incorporate different combinations of parameters representing these potential effects.

If we are also interested in the nature of each effect (assuming it exists), we can look at the parameter estimates associated with each factor. These parameter estimates can be challenging to interpret for a Bayesian ANOVA because unlike in Bayesian regression, JASP constrains the parameter values such that parameters representing different levels within the same factor must add up to 0. The grand mean of all the data acts as the intercept, and each effect is a deviation from the intercept. As we consider more complicated models, those deviations are combined additively to determine how parameter estimates play out in terms of specific combinations of factors. Even though parameter estimates are not our focus in the present example, we discuss them more thoroughly in the "Inferential Output" section and with concrete examples in the "Worked Example" section.

[15]Note that, based on the principle of marginality (Nelder, 1977), it is potentially problematic to consider models that account for interactions without accounting for the constituent main effects.

Specifying priors

In a Bayesian ANOVA, as in the Bayesian regression, we have two tasks. One is to determine which model best explains the data; the other is to make inferences about the effect sizes of the factors or interactions within each model. As such, we need two sets of priors.

Our default approach is again to use uninformed priors, indicating that we have no predictions that are specific to the present experiment. When it comes to determining which model best explains the data, we simply say that our prior expectations are that each model is an equally likely candidate. In the present example, we are comparing 5 models, so our priors are that each has a 0.2 prior probability. If we have a reason to favor some models over others *a priori*, JASP allows us to specify other priors by using the menu displayed in Figure 17.

When it comes to priors for the effect sizes of factors and interactions within those models, JASP's default is intended to be similar to what we were already doing with t-tests in the first case study. With t-tests, JASP defaults to a Cauchy distribution with a location of 0 and a scale of 0.707. Such a prior corresponds to the idea that small effects are generally more likely than large effects. For an ANOVA, we need to expand this prior to account for multiple dimensions, since we are considering multiple possible effects. Based on work by Rouder et al. (Rouder et al., 2012), JASP uses a multivariate Cauchy distribution, with an additional dimension for each effect under consideration.

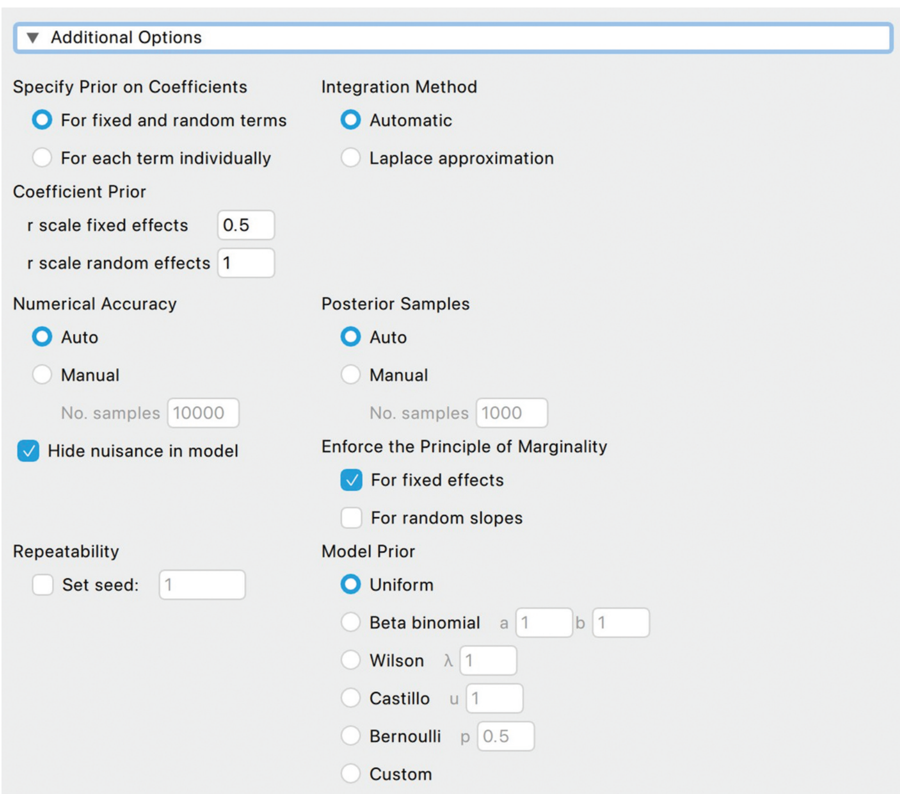


Figure 17. JASP menu for customizing ANOVA priors.

As before, the default priors are perfectly acceptable for most purposes. JASP defaults to a scale parameter of 0.5 for fixed effects and 1 for random effects, indicating that we assume large effect sizes are generally more likely for random effects than for the fixed effects of factors. However, the present example involves a between-subjects experiment where each data point is statistically independent, so we are not considering any random effects.

If these priors do not accurately reflect our expectations about plausible effect sizes, we can use the menu displayed in Figure 17 to adjust our priors. In particular, we can adjust how much credence the priors lend to the idea that small effect sizes are more likely than large effect sizes. For example, if we think large effect sizes are more plausible in a given experiment, we can increase the scale parameter. We can do so for all fixed and random effects if we think they are generally more likely to be large or small, or we can adjust priors one at a time for individual effects.[16]

Inferential output

As always, our inferential outputs focus on either testing our hypotheses about the existence of effects or on determining the size and direction of any potential effects. As in any Bayesian analysis, we use Bayes factors to summarize the evidence for or against a given effect, and we use parameter estimates to examine the size and direction of an effect. Parameter estimates are summarized with a 95% credible interval so as to get a sense of the range of reasonable values. In the present example however, our research questions are best answered by hypothesis testing, so we primarily focus on Bayes factors.

Since Bayes factors quantify the weight of evidence in favor of one hypothesis and against another, we can easily apply Bayes factors to compare two models. However, they become more complicated when we compare more than two models. As discussed in the "Bayesian Linear Regression" case study, we need to select a single model to act as a reference point. We recommend keeping JASP's default of using the best performing model as the reference point for all model comparisons. This recommendation is based on convenience rather than necessity. As long as we compare any single reference model to every other model, we can use that information to determine how any two models would compare. Yet we are typically most interested in comparing the best performing model to theoretically important alternatives, so we might as well use the best performing model as our reference point. For similar reasons, we recommend asking JASP to produce the default BF_{10} so that larger Bayes factors indicate better performing models. The best performing model will always have a (BF_{10}) of 1 (i.e., it is no better or worse than itself), and the other models will have decimal (BF_{10})s, which will keep getting smaller to indicate how much *worse* a given model is than the best performing model. Researchers will need to take the inverse of these values to determine how much *better* the best model is in comparison to a given model. We still suggest using this default because other options are liable to cause greater confusion.

If we are interested in the size and direction of any effects, we need parameter estimates. As with Bayesian regression, we need to be mindful of two forms of uncertainty: uncertainty about which model is likeliest and uncertainty about the values of parameters within each model. In our worked example for multiple regression, we demonstrated how the value of a given parameter can depend on what other parameters are included in the model. In that case, there was partial

[16]Note that selecting "For each term individually" opens a new menu that might only be visible by scrolling down.

confounding. When there is no confounding to worry about and the effects must be independent, we have one parameter estimate for models that include the parameter, and another parameter estimate —implicitly 0—for models that do not include it. Including a parameter in a model instantiates the hypothesis that the effect represented by the parameter does exist; excluding the parameter instantiates the hypothesis that it does not. JASP accounts for these competing parameter estimates by using model averaging. This procedure involves creating a weighted average of the parameter estimates from all possible models, where the weight given to a parameter estimate from a given model is based on the weight of evidence in favor of that model. Thus, in an experimental ANOVA, model averaging is essentially down-weighting our parameter estimate in proportion to the likelihood that the effect it represents does not actually exist.

Model-averaged estimates might be useful if we need to take action and want to hedge our bets. Yet in an inferential context, the procedure can seem somewhat odd in that it is accounting for the possibility that the effect may or may not exist by reducing our estimate for the size of the effect. This information may be useful if we need to take action based on our data and would like to proportionally handicap our actions. If we would instead prefer to separate our inferences about whether an effect exists from our inferences about its direction and magnitude, we can have JASP provide the estimates on individual models by using the "Single Model Inference" menu (Figure 18). However, our present example depends more on hypothesis testing than parameter estimation, so we will not go through the process of determining model-specific estimates here.

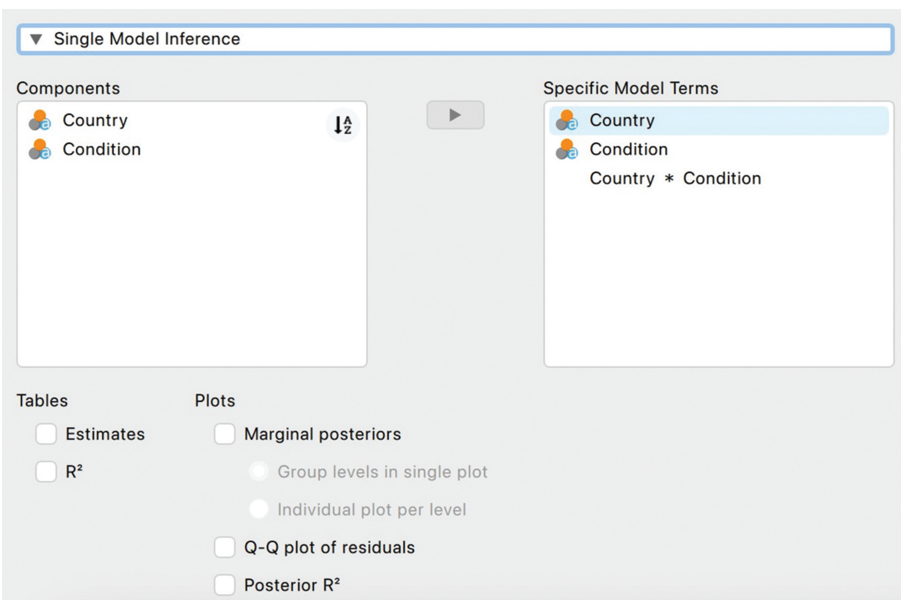


Figure 18. JASP menu for making inferences about individual models in a Bayesian ANOVA.

Worked example

We first load our data set and make sure that JASP is assigning the appropriate type of scale to each column (see Figure 1). Once we have our data loaded, we click on the ANOVA module (A in Figure 19) and select the Bayesian ANOVA.

Bayesian ANOVA

Once we open the ANOVA module, we need to select the variables that we will use in our analysis. JASP provides the options from the data we have loaded in Box B (Figure 19). In this case, our dependent variable is the “Imitation Score” and our independent variables are “Country” and “Condition.” Since both condition and country are fixed factors, we move those labels into the appropriate box (C in Figure 19) either by dragging them or by highlighting them and using the arrow buttons.

To examine or adjust our priors, we can use the “Additional Options” menu (Figure 17). We can change our priors for all effects in general or we can change our priors for specific effects. However, JASP’s default priors are sufficient for our purposes. We want to make minimal theoretical commitments and generally expect small effects to be more likely than large effects.

Having set up our data and our priors, the next step is to determine what outputs we want. We first select whether we would like Bayes factors reported in terms of evidence for

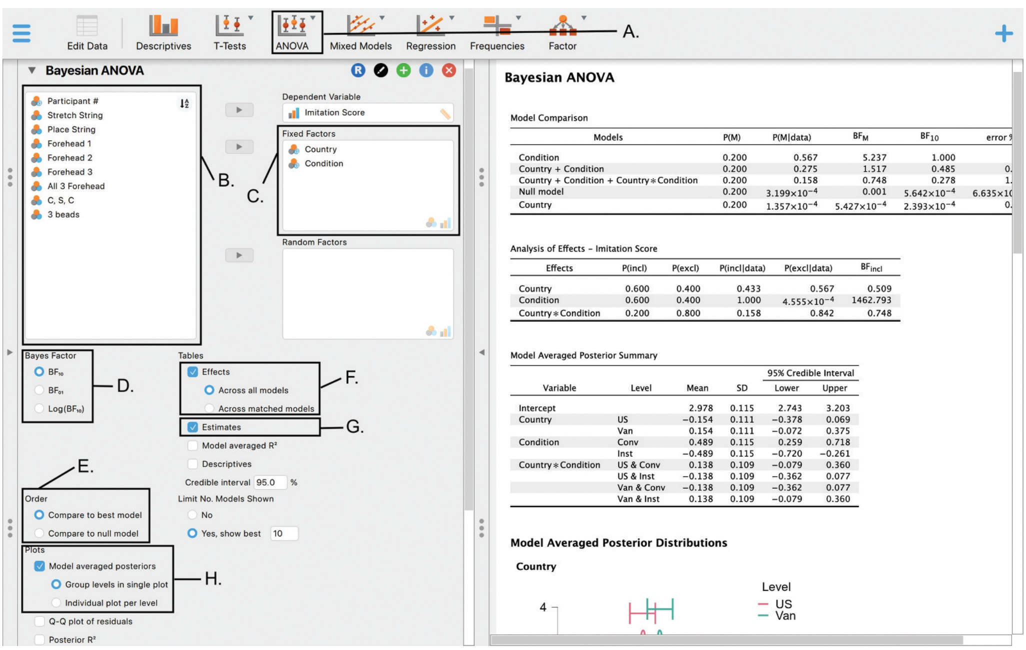


Figure 19. Main JASP menu for bayesian ANOVA. Note that it is more convenient when using JASP if we arrange data for between-subjects conditions in a .csv file that has the dependent variable in a single column, with separate columns identifying the precise condition from which each data point originated (i.e., “long” format). In this case, we have a single column for all participants’ imitation scores. We have additional columns identifying the condition to which a given participant was assigned and the country from which they were recruited.

the null hypothesis (BF_{01}) or for the alternative hypothesis (BF_{10}) (D in Figure 19). This choice does not alter our findings, but as with other situations where we are comparing multiple models (e.g., multiple regression), we recommend keeping the default (BF_{10}). We then need to indicate whether we would like JASP to use the null model or the best performing model as the reference point for pairwise model comparisons (E in Figure 19). We recommend "Compare to best model," which JASP does by default.

Next, we need to decide how we would like JASP to report the results of the model comparisons. By default, JASP provides a "Model Comparison" table, showing the relative performance of each possible model. We also have the option of requesting an "Effects" table (F in Figure 19), which essentially repackages the information from the model comparison table to show the overall evidence in favor of each effect. To quantify such evidence, JASP compares models that account for the effect with those that do not, providing two options for doing so. We can either compare all models that account for the effect to all those that do not or we can compare paired models where the only difference is the presence or absence of the predictor.[17]

To get information about the nature of each effect, assuming it exists, we can ask JASP to create an "Estimates" table (G in Figure 19). The contents of this table are calculated via model averaging. This procedure accounts for both our uncertainty about the existence and potential size of effects by creating weighted averages of posterior estimates, with model inferences weighted in proportion to how well each model has performed. If we want estimates based only on specific models, we can use the options provided in the "Single Model Inference" menu (Figure 18).

JASP can also produce useful plots. Beyond plots that check the assumptions involved in conducting an ANOVA in the first place, we are probably most interested in visualizing potential effects. To do so, we can select the "Model averaged posteriors" option under the "Plots" heading (H in Figure 19). The resulting plots (Figure 20) are visualizations of the information provided in the "Estimates" table. They show us what we now believe about those effects, given our prior expectations and what we have learned from our data. Additional options for descriptive plots can be found under the "Descriptive Plots" and "Raincloud Plots" menus.

Interpreting and reporting results

The first output JASP provides is the model comparison table (Figure 21). This table allows us to determine the extent to which various proposed effects are supported by the data. The first column tells us what predictors are included in the model. The "P(M)" column displays our priors regarding how likely each model is. Since we used the default uniform distribution, these values are equal. The "P(M|data)" column shows our posterior inferences for how likely a given model is in light of the data. The "BF_M" column provides a Bayes factor quantifying how the odds have changed from our prior to posterior probability; values greater than 1 indicate that the model has become more likely in light of the data, whereas values less than 1 indicate the opposite. The final "Error" column refers to the approximations JASP uses in its calculations and is not typically a cause for concern.

[17]This choice relates to a fundamental debate about how model-based ANOVAs should approached in the first place (i.e., what models are fair and appropriate to consider; Heathcote & Matzke, 2021; van den Bergh et al., 2023; van Doorn et al., 2023), but the difference in results is generally minor.

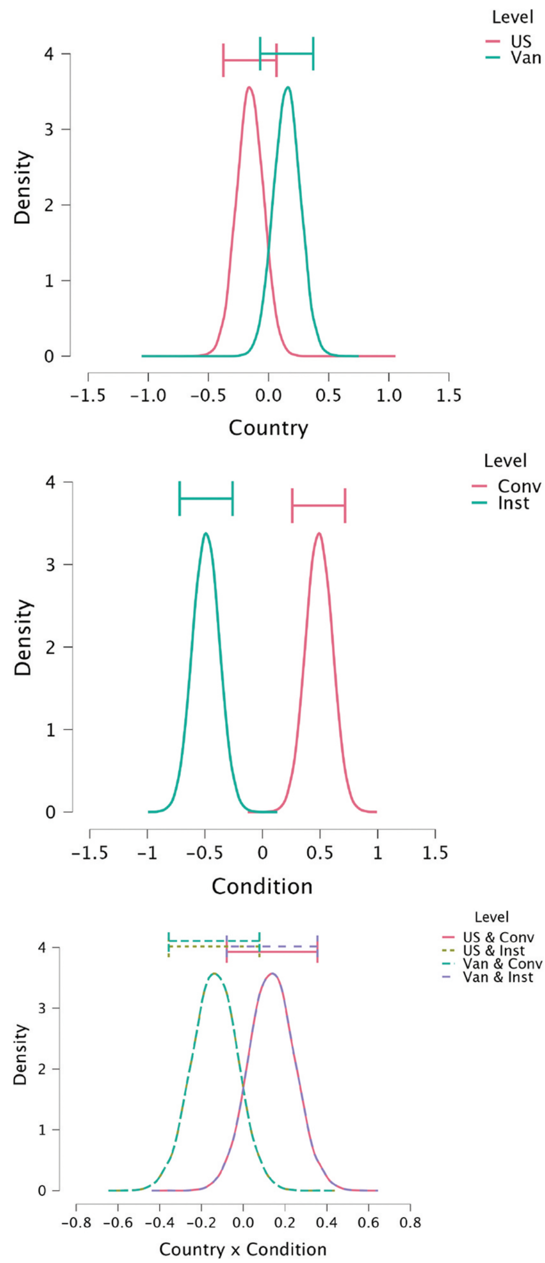


Figure 20. Posterior distributions for main effect of country (top), main effect of condition (middle), and interactions between condition and country (bottom).

Of primary interest to us is the "BF_{10}" column, in which we can see the results of pairwise model comparisons. This column tells us how much better a given model performs when compared to our reference model. In the preceding section, we advocated using the best model as our reference model. Thus we see that the best model, which only accounts for condition, has a BF_{10} of 1, as it performs no better or worse than itself. In contrast we see that the null model has a BF_{10} of $5.64 * 10^{-4}$ against it relative to the reference model. To

Model Comparison

Models	P(M)	P(M|data)	BF_M	BF_{10}	error %
Condition	0.200	0.567	5.237	1.000	
Country + Condition	0.200	0.275	1.517	0.485	0.815
Country + Condition + Country ∗ Condition	0.200	0.158	0.748	0.278	1.104
Null model	0.200	3.199×10^{-4}	0.001	5.642×10^{-4}	6.635×10^{-10}
Country	0.200	1.357×10^{-4}	5.427×10^{-4}	2.393×10^{-4}	0.026

Figure 21. Table summarizing model comparison results for Bayesian ANOVA.

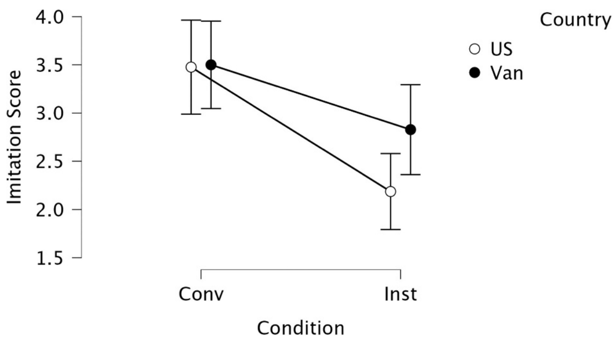


Figure 22. Main JASP menu for bayesian ANOVA.

make this result easier to interpret, we can take the inverse of BF_{10}, which gives a Bayes factor of 1/0.000564 = 1772 in favor of the model the condition-only model relative to the null model. Since a model that accounts for condition performs much better than a model that does not, we have strong evidence in favor of a main effect of participants observing a conventional vs. instrumental demonstration.

When we look at models that also account for country and an interaction between country and condition, the evidence is less definitive. We see that the model which accounts for both condition and country has a BF_{10} of 0.485. We can take the inverse to find a Bayes factor of approximately 2 in favor of the condition-only model. Similarly, the full model has a BF_{10} of 0.278. We can take the inverse to find a Bayes factor of approximately 3.5 in favor of the condition-only model. We would typically classify both of these Bayes factors as constituting weak or anecdotal evidence, so we would not want to draw firm conclusions as to whether participants' country has any effect beyond the clearly demonstrated effect of condition.

Looking at a descriptive plot (Figure 22), these model comparison results make sense. At first glance, we see what looks like a possible interaction effect. Participants in Vanuatu and the US have virtually identical mean scores in the conventional condition but somewhat different mean scores in the instrumental condition. We can describe the nature of any effects numerically by examining parameter estimates. The "Intercept" of 2.978 represents the grand mean, and effects are additive deviations away from that grand mean.[18]

[18]Note that the posterior distributions for levels within single factor are symmetrical around 0 because of an assumption built into the models, namely that those levels must sum to 0. For example, what would we expect the imitation score to be for a US participant in the instrumental condition, if we assume that there is an interaction between condition and country? We first start with the grand mean of 2.978. We decrease our estimate by 0.154 due to main effect of condition, we decrease our estimate by another 0.489 due to the main effect of country, and we decrease our estimate by an additional 0.138 due to the interaction between country and condition. Thus in a model that assumes an interaction, we would expect a US participant in the instrumental condition to have an imitation score of 2.197.

When we move beyond point estimates and consider our uncertainty about those estimates, we see overlapping error bars in the instrumental condition. These error bars represent 95% credible intervals based on our posterior distributions for each parameter estimate. Since the 95% credible intervals have a range of reasonably likely values that partially overlap, we cannot draw strong conclusions for or against there being differences between the two research sites. Having seen our data, it is quite reasonable to suspect that there is no difference, and it is also still somewhat reasonable to suspect that there is a difference.

For this analysis, we might report the following:

> Past research indicates that children imitate an adult's demonstration more closely if they believe the adult is demonstrating how to follow a tradition rather than how to achieve a goal. In the present experiment, we examine whether this effect is consistent across cultures by recruiting participants in the US and in Vanuatu. Each participant observed a demonstration of how to make a necklace. Participants were either told that the demonstration was goal-oriented (instrumental condition) or representative of how people typically accomplish the task (conventional condition). Participants were then asked to make a necklace themselves. Based on how they went about making the necklace, they were given an imitation fidelity score ranging from 0 to 5, corresponding to how many of 5 key actions were imitated.
>
> To analyse this 2×2 between-subjects experiment, we conducted a Bayesian ANOVA in JASP (JASP Team, 2022), with both country and condition as fixed factors. We used JASP's default uninformed priors for both model comparison and for parameters within each model. We thus specified our priors as (1) a uniform distribution across all possible models and (2) a multivariate Cauchy distribution for parameters within the models, with a location of 0 and a scale of 0.5. The first component of this prior indicates that we have no reason to expect any model to outperform any other model before having seen the data. The second component of this prior indicates that we generally believe small effect sizes are more likely than large effect sizes.
>
> Based on model comparisons, we find very strong evidence for a main effect of condition (BF_{10} = 1772), replicating past research indicating a difference between how faithfully children imitate an adult's demonstration in the conventional condition and the instrumental condition. By looking at the parameter estimates, we can see that, as with past research, the difference is caused by children imitating adults with greater fidelity in the conventional condition (±0.489, 95% CI [0.259, 0.718]). Our results are less clear when it comes to differences between the countries. We find very weak evidence in favor of there being no main effect of country (BF_{01} = 2), and similarly weak evidence in favor of there being no interaction between condition and country (BF_{01} = 3.5). This ambiguity makes sense in the context of a descriptive plot of our data (Figure 22). We do see a larger difference between US and Vanuatu for the instrumental condition than we do for the conventional condition, but we also see considerable overlap in the 95% credible intervals.
>
> In summary, our results constitute a successful replication of past findings that children imitate demonstrations of tradition more faithfully than goal-oriented demonstrations. Our results are ambiguous as to whether this effect varies by country. We find tentative evidence in favor of there being no differences between the research sites, but further cross-cultural research is needed to clarify the situation.

General discussion

Our goal in this tutorial has been to outline relatively simple ways in which developmental researchers can begin to employ Bayesian methods. To do so, we have laid out 5 case studies demonstrating how some of the most common statistical tests can be adapted to align with a Bayesian conception of probability. In looking back at these case studies, we can identify common benefits and challenges to such an approach.

The first challenge that we consistently encounter in these Bayesian analyses is specifying our prior expectations. Priors can be particularly disorienting for researchers who are new to Bayesian methods and thus are unaccustomed to incorporating background information as an explicit component of a statistical analysis. Frequentist statistical tests have been routinized such that they can be broadly applied. Creating comparable Bayesian routines means adopting priors that are relatively noncommittal such that the testing procedures can applied just as broadly. But as illustrated in the above case studies, we inevitably end up making some commitments in specifying our priors. Indeed, a complete lack of commitments is itself a commitment—and one that is often difficult to justify. Instead, we focus on making as few commitments as possible and ensuring that those commitments are acceptable given what we know about the situation being analyzed. Even then, it is still quite difficult to create generally acceptable priors that make minimal commitments. As such, considerable research has gone into developing the uninformed priors that JASP uses as a default (Wagenmakers et al., 2018).

In addition to posing a challenge, priors can also be beneficial because they encourage researchers to think about the context in which data are collected and analyzed. Researchers often view statistical procedures as objective, but in practice the results depend on the decisions and assumptions researchers make both in collecting and analyzing data (Chambers, 2017; Westfall et al., 2016). Priors can provide a framework for being explicit about our expectations and the background information on which those expectations are based. This context also becomes important for how we interpret our results.

To interpret our results from each of these analyses, we are always considering two types of inferential outputs. Our first type of output is the Bayes factor. Since Bayes factors allow us to quantify the relative evidence for two competing hypotheses, they are our primary inferential output for determining whether a given effect exists. One of the two competing hypotheses is often a null hypothesis, so Bayes factors also allow us to quantify evidence against an effect existing (i.e., in favor of the null), which can be difficult to do with frequentist tests (Dienes, 2014). But Bayes factors are not limited to the framework of traditional null hypothesis testing. Indeed, they can compare any two hypotheses that have been translated into statements about model parameters (Dienes, 2008; Etz et al., 2018). There are also other ways to compare models that take different approaches to quantifying the trade-off between model complexity and model fit (e.g., Gelman et al., 2020, and cited works therein).

How much evidence do we need before we accept a hypothesis? Rigid thresholds are not typically the best approach. Bayes factors quantify the weight of evidence on a continuum; what we should conclude based on that evidence depends on the context. If we are looking for evidence of a subtle effect in a naturalistic setting, we may settle for relatively weak evidence. On the other hand, if we are looking for evidence of a psychophysical law in a tightly controlled experimental setting, we might demand a much stronger Bayes factor.

Our second type of inferential output is the credible interval. Credible intervals provide information about the likely direction and magnitude of parameters. In a Bayesian framework, our parameter estimates are based on posterior probability distributions that describe the relative plausibility of possible values. We typically report 95% credible intervals to summarize our uncertainty about the "true" parameter value. Since credible intervals are merely a summary of the posterior distribution, we should be cautious about over-interpreting their endpoints.

Credible intervals often corroborate Bayes factors because the competing hypotheses make statements about our parameter(s) of interest. A typical null hypothesis is that the true value of a parameter of interest is 0, with a corresponding alternative hypothesis being that the true value is something else. When the data suggest that parameter values close to 0 are plausible, the Bayes factor will tend to favor the null hypothesis. However, unlike the one-to-one relationship between p-values and confidence intervals, there is no direct correspondence between the Bayes factor and credible interval. Bayes factors test whether an effect exists, and credible intervals inform us about its likely direction and magnitude. While these two outputs are typically complementary, there may be edge cases where the conclusions we would draw from the Bayes factor do not obviously align with those we would draw from credible intervals.

In establishing parameter estimates for statistical tests that involve more than two models (e.g., linear regression, ANOVA), we are faced with the decision of whether to employ model averaging. At its core, this decision hinges on whether our research question is best answered by focusing exclusively on individual models, in which case we would avoid model averaging. In the linear regression case study, we focused on estimates from individual models because we wanted to compare parameter estimates from two models to investigate potential confounding. In the ANOVA case study, we largely avoided the issue because our research question was best answered via hypothesis testing. Thus we do not use model averaged estimates in either of our worked examples. There are situations in which model averaging would be preferable (Etz et al., 2016; Gronau et al., 2021; Guan & Vandekerckhove, 2016). Yet if we have strong evidence, our decision regarding model averaging will not qualitatively alter our conclusions. If we lack strong evidence, we cannot draw firm conclusions in any case.

Our priority in this tutorial has been to present Bayesian methods in a way that lowers the barrier to entry for developmental researchers. As such, we have described Bayesian adaptations of the most widely-used statistical routines. We have done this to make it easier for new users of the methods to map the information onto what they already know. However, Bayesian modeling can be used to do much more than this. It is a broad and diverse framework for drawing inferences from data. When used to its full potential, this framework can allow researchers to tailor their statistical analysis to fit their research questions rather than tailoring their research questions to fit their statistical analysis (Bürkner, 2017; Lee & Wagenmakers, 2014). We have not delved into bespoke Bayesian modeling here, because we recognize that it may seem daunting to those who have been trained exclusively in frequentist methods. We have instead focused on how researchers can gain many of the benefits of adopting a Bayesian approach, even within the somewhat unnatural re- strictions of null hypothesis testing routines. In doing so, we hope to provide developmental researchers with an accessible and immediately applicable introduction to using Bayesian methods, which can be a jumping-off point for further growth and exploration into these flexible and intuitive techniques.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The JASP files and archival data used are available in the Open Science Framework repository https://osf.io/ta9re/.

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